ConfabQA: Hidden-State Refusal and Correctness Signals in Three Small Language Models

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2026-08-15

Abstract

When a small language model gives a confidently wrong answer (a *confabulation*), does its hidden state at the moment of commitment already encode the warning sign? A standard probing experiment fits a linear classifier, a *probe*, on the model's hidden states and reports correctness-prediction accuracy in the 80–90% range. The reported result carries two structural confounds. First, correctness is confounded with the knowledge cutoff: the benchmarks the probe is typically trained on (PopQA, TriviaQA, and similar factual-QA sets) pair pre-cutoff items the model gets right with post-cutoff items the model gets wrong, so a probe predicting "correct" is mathematically indistinguishable from one predicting "this question is pre-cutoff." The probe might be a date detector, not a knowledge detector. Second, hidden-state signal is confounded with prompt features: a logistic regression trained on the question text alone often reaches comparable accuracy on the same labels, and a hidden-state probe that does not beat that simpler baseline has extracted no model-internal information at all. The probe might just be classifying inputs. Without controls for both confounds, the headline probe accuracy may not reflect what the model knows about its own answer.

ConfabQA is a 784-item probing benchmark: 4 domains $\times$ 3 categories (well-known pre-cutoff, *obscure* pre-cutoff, post-cutoff). The obscure category (questions whose answers are in the training data but which the model is expected to get wrong) breaks the cutoff confound. Each item carries a provenance URL and an external-LLM validation status.

The second confound is addressed in the protocol, not the dataset: every probe, on every model and dataset, is scored against the strongest of four prompt-feature classifiers fit on the same labels and folds, and only the margin over that baseline counts as model-internal signal. Applied to three instruction-tuned subject models (Qwen3-1.7B, Gemma 2 2B, Llama 3.2 3B), with Qwen3-4B as a within-family scaling control, each subject model carries a *refusal* direction in its late layers, while the *correct-vs-wrong* margin beyond prompt features is sharply model-dependent: large and dataset-general on Llama 3.2 3B, small and dataset-local on Qwen3-1.7B and Gemma 2 2B. A sparse-autoencoder decomposition of the Qwen3-1.7B refusal direction resolves it into a canonical refusal-opener feature, a dormant apology-opener feature, and two post-cutoff content-cue detectors; adding the opener feature alone is causally sufficient to flip the model's next-token argmax on wrong items from a confident answer to a refusal opener. The probe direction is a mixture of separable features, and a single one of them carries the causal effect.

Cross-dataset transfer answers the portability question for correctness directly: a probe trained on one benchmark retains 81% of its margin on unseen benchmarks for Llama 3.2 3B but collapses for Qwen3-1.7B and Gemma 2 2B. Three research questions remain open: whether the refusal direction transfers the same way, whether the cross-family gap survives at $\gtrsim$ 7B, and whether swapping the abstention-training recipe while holding the model family fixed reproduces it. I release ConfabQA, the judge, the full analysis code, and the cached hidden-state activations needed to tackle all three.

1. Introduction

A language model **confabulates** when it produces a fluent, confident answer that is factually wrong (the term follows Farquhar et al. 2024). Kadavath et al. (2022) argued that LLMs "mostly know what they know": signals internal to the model can predict whether a given answer will be correct. A probing literature has followed (Azaria and Mitchell 2023; Marks and Tegmark 2024), fitting linear classifiers on model hidden states and reporting correctness-prediction accuracies in the 80–

90% range. Behavioral detectors, such as sampling consistency (Manakul et al. 2023) or semantic entropy (Farquhar et al. 2024), work on the outputs instead and are the black-box complement to the white-box probes studied here. Two concurrent 2026 studies push back: Singh et al. show that above-chance "introspection" probe accuracy on small LMs is explained by features of the prompt itself rather than privileged self-access; Sahoo et al. find the same pattern for reasoning-mode probes. The question is no longer whether the probe's numbers are high, but whether they are *evidence*.

This paper is a comprehensive small-LM test of that question. I introduce **ConfabQA**, a probing benchmark structured as 4 domains $\times$ 3 categories (well-known pre-cutoff, *obscure* pre-cutoff, post-cutoff). The third category (items whose answers existed in the model's training data but on which the model is expected to fail) populates the cell that standard pre-vs-post-cutoff designs leave empty, breaking the cutoff/correctness confound that lets a probe predicting "correct" double as a probe predicting "pre-cutoff date." ConfabQA, plus 800-item samples of PopQA (Mallen et al. 2023) and TriviaQA (Joshi et al. 2017), are run through three instruction-tuned models in the 1.7–3B range (Qwen3-1.7B, Gemma 2 2B, Llama 3.2 3B), with Qwen3-4B as a within-family scaling control. Every probe is reported against both the standard majority baseline and a stricter logistic-regression-on-question-text baseline: a control rarely applied in the probing literature, but the one that decides whether the hidden state contributed anything.

Linear probing as a tool. The diagnostic-classifier framework was introduced by Alain and Bengio (2017): fit a linear classifier on the hidden representation at each layer of a trained network to test whether a property is linearly decodable. Subsequent interpretability work (Belrose et al. 2023 on tuned lenses, Burns et al. 2023 on contrast-consistent probes) has applied probing to large LMs specifically. The present work uses the same machinery but with a prompt-feature control baseline. This is a *selectivity control* in the lineage of Hewitt and Liang (2019), who showed that probe accuracy without a control task mostly measures the probe rather than the representation; Belinkov (2022) surveys the same failure modes. Controls of this kind remain rare in the correctness-probing literature.

Knowledge cutoff and the disconfound. Several prior probing papers use time-sensitive question sets and conflate "post-cutoff" with "model doesn't know." The ConfabQA 4×3 design (§3) is motivated by exactly this conflation; the prompt-feature baseline (§4) is motivated by the observation that even after the cutoff confound is removed, *some* feature of the question text still trivially predicts correctness on small LMs.

Closest mechanistic analogue. Ferrando et al. (ICLR 2025) identify *entity-recognition latents* in Gemma-2 SAEs that causally steer the model between knowledge-grounded answers and refusals on factual questions, similar to the Qwen3-1.7B experiment here (§6.3). The present work extends that line in two ways: (i) it recovers the same class of feature via a linear probe + SAE pipeline on a different model family with a 4×3 disconfounded benchmark; and (ii) it adds the cross-dataset transfer experiment (§8.3) that asks whether the recovered direction is *content-invariant* (as the entity-recognition framing suggests) or dataset-local.

Findings. Two complementary results emerge from the comprehensive read that no single-model, single-dataset study could:

1. *Refusal vs. confabulation is a model-internal feature: each model linearly encodes its own upcoming speech act.* On the wrong+refusal subset of ConfabQA the late-layer probe beats the strongest prompt baseline on every model: +7.4 pp on Qwen3-1.7B, +16.3 on Gemma 2 2B, +2.2 on Llama 3.2 3B. The less predictable a model's refusals are from the question text alone, the more the probe adds (Section 7.2). That some such signal exists is natural: the refusal-vs-answer label is the model's own output decision, and the late hidden state is the computation that produces it. The findings are the signal's *form*: a single linear direction at the deepest layers, causally sufficient, and decomposable. A one-shot residual-stream intervention drives the first-token refusal-opener rate to 100% on Qwen3 and Gemma (Sections 6.2 and 7.2). This replicates Arditi et al.'s (2024) single-direction refusal finding on three model families, with the cutoff variable controlled by construction. The regime differs: theirs is safety refusal on harmful prompts, here it is epistemic abstention on unknown-answer questions. The direction is not, however, a single feature. A sparse-autoencoder decomposition resolves it into a canonical refusal-opener feature, a dormant apology-opener alternative, and

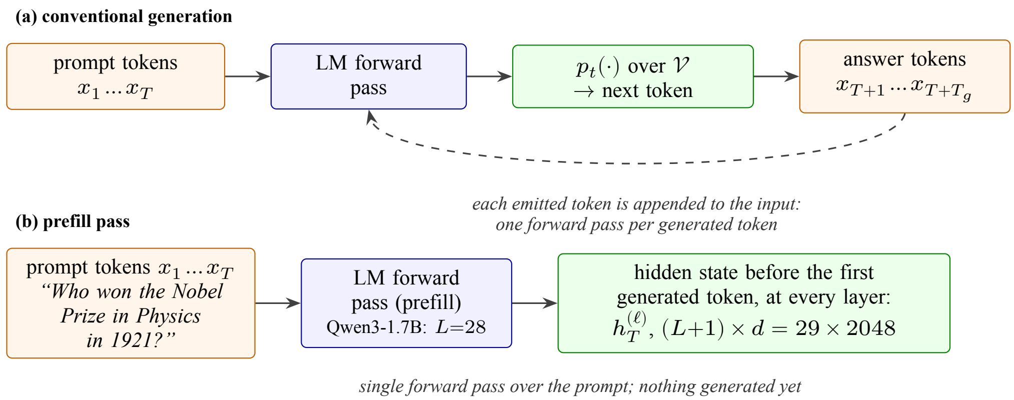

Figure 1: Token flow in the two forward-pass regimes. **(a)** Conventional generation runs one forward pass per generated token, feeding each emitted token back into the input. **(b)** The prefill pass stops just before the first generated token; the hidden state at that moment, $h_T^{(\ell)}$, is saved at every layer ℓ and is what the probes read. Generation starts from exactly this state, so it is the model's representation immediately before it commits to an answer; stacked over questions it forms the probe input tensor $(n,\ L{+}1,\ d) = (784,\ 29,\ 2048)$. The dimensions ($L$=28, d=2048) are specific to Qwen3-1.7B; the other subject models differ in both depth and width.

two content-cue detectors (Section 6.3). The opener feature alone is causally sufficient to flip $30/30$ wrong-item next-token argmaxes to refusal openers.

2. *Llama 3.2 3B carries a substantially larger and more general correctness signal than Qwen3-1.7B or Gemma 2 2B.* On balanced PopQA / TriviaQA subsamples, Llama's probe beats the prompt baseline by $+21$–$+25$ pp, $5.7\times$ and $2.2\times$ Qwen's margins on the same benchmarks. Three controls localize the source: it is not parameter count (a Qwen3-4B within-family scaling control moves the margin by less than 2 pp; Section 8.1); it is not dataset-specific (Llama's probe keeps $\approx 81\%$ of its margin when transferred across datasets without refit, where Qwen's and Gemma's transfer collapses; Section 8.3); and it is not refusal-channel readout ($\approx 83\%$ of the margin survives dropping refusals, and an independent refusal probe adds a further $+18$–$+25$ pp gap; Section 8.2). Gemma is intermediate in behavior (refusal rate and refusal-probe margin between the other two), yet its correctness probe is the most dataset-local of the three. The most plausible single explanation: Llama's heavy calibrated-abstention training forces a content-invariant "model is uncertain" representation; neither Qwen3's nor Gemma's training developed it.

The contribution is a comprehensive small-LM probing study with the prompt-feature baseline applied uniformly: a curated benchmark whose design controls for the cutoff confound, a cross-dataset bootstrap that exposes the cross-model gap, a refusal-channel attribution that partitions the gap, an SAE decomposition that resolves the refusal direction into interpretable features with causal validation, and a cross-dataset transfer experiment that asks whether the recovered correctness signal is universal or local to its training set.

2. Theoretical background

The subject models are decoder-only transformers with causal attention. The model reads the prompt $x_{1:T}$ as a token sequence; in every layer, each position attends only to positions at or before it (the causal mask), so the state at position t is a function of $x_{1:t}$ alone. The final-layer state at the last position is projected through the language-model (LM) head, the output projection onto the vocabulary $\mathcal{V}$, into a probability distribution over the next token, and generation repeats this one token at a time, feeding each emitted token back as input (Figure 1a). The *prefill* pass (Figure 1b) is the single forward pass over the prompt alone, before any token is generated; it is the pass the probes of this paper read. Qwen3-1.7B instantiates this with $L = 28$ blocks, hidden size $d = 2048$, and a vocabulary of $|\mathcal{V}| \approx 152{,}000$ tokens with tied input/output embeddings (Methods §4 lists the checkpoint and settings). This section defines the two confidence metrics used in §5.2 (§2.1), the prefill hidden state the probes read (§2.2), the linear-probe pipeline (§2.3), and the two confounds that motivate the dataset and baseline design (§2.4).

2.1 Confidence metrics: log-probability and entropy

At each generation step t the model produces a probability distribution $p_t(\cdot)$ over the next token. All generation in this paper is greedy (`do_sample=False`, equivalent to temperature 0, hence deterministic), so the emitted token is the mode of p_t. Two scalar summaries of the generated response recur in §5, both means over the T_g generated positions ($t = T, \dots, T+T_g-1$, with T the prompt length and $T_g \leq 64$):

$$\bar{\ell} = \frac{1}{T_g} \sum_{t=T}^{T+T_g-1} \ell_t, \qquad \ell_t = \log \max_{x \in \mathcal{V}} p_t(x), \tag{1}$$

$$\bar{H} = \frac{1}{T_g} \sum_{t=T}^{T+T_g-1} H_t, \qquad H_t = -\sum_{x \in \mathcal{V}} p_t(x) \log p_t(x) = \mathbb{E}_{x \sim p_t}[-\log p_t(x)]. \tag{2}$$

Mean per-token log-probability $\bar{\ell}$. In (1), ℓ_t is the log-probability of the token emitted at step t (the argmax, under greedy decoding). The log, rather than the raw probability, makes the per-step values additive: $T_g\bar{\ell}$ is the log-probability of the entire generated answer, so $\bar{\ell}$ is the length-normalized answer likelihood. Equivalently, $\exp(\bar{\ell})$ is the geometric mean of the per-token probabilities, so $\bar{\ell} = -0.05$ corresponds to ≈ 0.95 per token. $\bar{\ell}$ close to 0 means every step was near-certain; more negative means some steps were close ties. I also report $\min_t \ell_t$ as a worst-step diagnostic.

Mean per-token entropy $\bar{H}$. In (2), H_t is the Shannon entropy of p_t in nats; the second form reads it as the expected surprisal of a token drawn from p_t. It measures the shape of the whole distribution, independent of which token was chosen: a near-deterministic p_t has $H_t \approx 0$, a uniform one has $H_t = \log|\mathcal{V}|$.

$\bar{\ell}$ and $\bar{H}$ measure different things. ℓ_t looks at one token, the mode of p_t; H_t is a probability-weighted average of $-\log p_t(x)$ over all tokens. So $\bar{\ell}$ asks "how confident was the model in the token it picked," while $\bar{H}$ asks "how concentrated was the whole distribution." The two are tightly anti-correlated empirically (Figure 5b, §5.2) but can come apart: one dominant mass plus a long thin tail of plausible alternatives gives large H_t (many small probabilities, each multiplied by a large $-\log p$) while ℓ_t stays close to 0.

2.2 The prefill hidden state as the canonical going-in representation

When the model is asked a question, the *prefill* pass (a single forward pass over the prompt $x_{1:T}$) produces the full hidden-state tensor $h_t^{(\ell)}$ for $\ell = 0, \dots, L$ and $t = 1, \dots, T$. I pick off the **hidden state at every layer just before the first generated token** (the last prompt position), $\{h_T^{(\ell)}\}_{\ell=0}^{L}$, yielding a tensor of shape $(L+1, d) = (29, 2048)$ per question. Here T is the final token of the *question itself*: nothing has been generated yet, so $\{h_T^{(\ell)}\}$ is the model's state at the moment it is about to emit the first answer token: its representation *immediately before it commits to the answer* (Figure 1b). Three reasons this is the natural probe target:

1. **Causal masking.** Because attention is causal, $h_T^{(\ell)}$ summarizes the whole prompt seen so far; no later prompt-position state contains additional information about the prompt.
2. **Output-locality at the final layer.** $h_T^{(L)}$ is, up to RMSNorm and a linear projection, the first answer token's logit vector. A property linearly decodable from $h_T^{(L)}$ is, by construction, available to the model's output head at the moment of commitment.
3. **Avoidance of generation contamination.** Probing the hidden state of *generated* tokens conflates "what the model knows" with "what the model has already committed to saying," because the generated prefix is itself fed back as input. The prefill state at T is the cleanest pre-commitment representation.

2.3 Why a linear probe (and the specific pipeline I use)

Following Alain and Bengio (2017), I fit at each layer ℓ a binary linear classifier on $\{h_T^{(\ell)}\}$ predicting a property of interest. A linear probe at layer ℓ achieving accuracy substantially above the appropriate baseline implies that there exists a hyperplane in $\mathbb{R}^d$ that separates the property in the representation

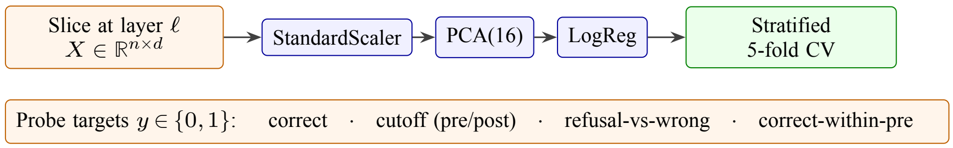

Figure 2: Per-layer linear-probe pipeline. At each layer ℓ, the $n \times d$ matrix of captured hidden states is passed through StandardScaler $\rightarrow$ PCA(16) $\rightarrow$ logistic regression and evaluated with stratified 5-fold CV, separately for each binary target.

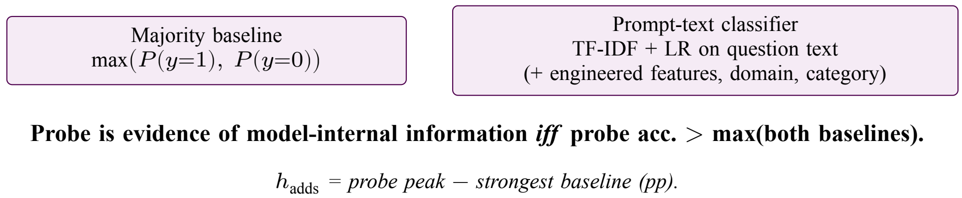

Figure 3: Baselines, fit on the same labels with no access to the hidden state: the majority class, and a prompt-text classifier, TF-IDF (term-frequency / inverse-document-frequency bag-of-words) plus logistic regression, with engineered features and domain/category dummies. A probe is evidence of model-internal information only insofar as it beats both; h_{adds} = probe peak − strongest baseline (pp), reported throughout §§5–8.

at depth ℓ. Equivalently: the linear span of the hidden state contains a direction along which the property is decodable. Whether the model *uses* that direction is a separate question.

Figure 2 shows the probe pipeline. With $n = 784$ items and $d = 2048$ features, training a raw logistic regression directly on the hidden states would massively overfit: with more free parameters than training items, the classifier can separate arbitrary labels without learning anything generalizable. The pipeline `StandardScaler -> PCA(n_components=16) -> LogisticRegression(max_iter=2000, C=1.0)` (the hyperparameter `C` is sklearn's inverse regularization strength, left at its default) projects onto the top 16 principal directions before fitting, keeping the ratio of training items to fitted parameters at $\approx 19{:}1$ instead of $\approx 0.15{:}1$. Both the scaler and PCA are fit only on the training fold (`sklearn.Pipeline` semantics) so no test-fold leakage occurs. 5-fold stratified CV (each fold preserves the class balance) with `random_state=0` produces five held-out evaluations per layer; I report the mean and per-fold std of accuracy. Appendix G sweeps $n_{\text{components}}$ over $\{4, 8, 16, 32, 48, 64\}$ to show the headline numbers are not specific to 16.

2.4 The two confounds and what counts as evidence

Confound 1 (cutoff/correctness). Answering wrong is naturally correlated with asking about post-cutoff facts. A benchmark built from easy pre-cutoff questions plus post-cutoff questions makes "the answer is correct" and "the question is pre-cutoff" nearly the same label, so a correctness classifier is, in effect, a cutoff classifier: the probe may be reading "this question mentions 2025", not "the model is about to be wrong". ConfabQA disconfounds by adding *obscure pre-cutoff* questions (facts that were in the training data but that the model still gets wrong) so that pre-cutoff no longer implies correct. Keeping only pre-cutoff questions would also remove the confound, but it would remove the refusals too: all of Qwen3-1.7B's refusals happen on post-cutoff questions (Section 5.1), so without the post-cutoff cell there would be no refusal behavior to study. The disconfound test is then: fit the correctness probe *restricted to pre-cutoff items only*; if its accuracy still exceeds the pre-cutoff majority baseline, the cutoff variable cannot account for the probe's signal.

Confound 2 (hidden state/prompt features). A correctness probe that beats its majority baseline is not yet evidence of *model-internal* self-knowledge. The probe might be reading features of the question text that any classifier could extract: the year mentioned in the question (which correlates with answerability), domain (some domains are easier), question length, presence of particular keywords. The honest baseline is therefore a logistic regression on prompt-side features alone, with no access to the model's hidden state. Section 4 specifies the exact feature set; Section 5.6 reports the comparison. A hidden-state probe is only evidence of model-internal information to the extent it beats this baseline by a margin outside per-fold noise. Figure 3 summarizes the two baselines.

This second baseline is the one the original probing literature most often omits. The cutoff disconfound is necessary but not sufficient; the prompt-feature disconfound is what actually distinguishes "the hidden state encodes self-knowledge" from "the hidden state encodes the prompt, which encodes the answer."

3. Sourcing and curation of the ConfabQA question set

ConfabQA comprises 784 items in a 4×3 design: 4 domains (science, history, culture, cinema) crossed with 3 categories (well-known pre-cutoff, obscure pre-cutoff, post-cutoff). Per-cell counts:

Table 1: ConfabQA cell counts: 4 domains $\times$ 3 categories, $n = 784$.

domain	well-known	obscure	post-cutoff	total
science	42	45	119	206
history	33	37	127	197
culture	34	34	121	189
cinema	34	37	121	192
total	**143**	**153**	**488**	**784**

Items are generated from per-domain templated source files; each item carries a provenance URL for its gold answer and a validation status field populated by an external LLM with web access. Three iterations of the validation pipeline were run on the source files; the final one, **ConfabQA** ($n = 784$), is what the paper analyzes throughout. The full design, schema, selection criteria, and known limitations are documented in `data/QUESTIONS_v1_CARD.md`. A fifth domain (sports) was included in an early iteration but produced $0/26$ correct across all three categories on this model, leaving no positive examples for the within-pre-cutoff probe; sports is therefore excluded from the main analysis and reported separately in Appendix D.

The third (`obscure`) category is the design innovation. It populates the cell "answer-was-in-training-data but model-is-expected-to-fail," which the standard two-cell design (pre-cutoff known-good vs. post-cutoff known-bad) leaves structurally empty. A typical obscure item: "Who won the Academy Award for Best Actor at the 1968 ceremony for 'In the Heat of the Night'?" (gold: Rod Steiger). The fact is on the public record, but Qwen3-1.7B confidently answers with the wrong actor. The model scores 33.3% on the obscure cell against 62.2% on well-known (Section 5.1).

The validation pipeline (`01_question_set.py --emit-validation-prompt` -> external LLM -> `--apply-validation`) makes the set reproducible: re-running it against the same source files and the same validation results yields a bitwise-identical question set, and gold corrections leave an audit trail in `validation_notes`.

4. Methods

Subject model. Qwen3-1.7B (`Qwen/Qwen3-1.7B`; Qwen Team, 2025a), a 1.72B-parameter text-only instruction-tuned decoder-only transformer (Apache 2.0). 28 transformer blocks, hidden dimension 2048. Loaded at bfloat16 on Apple Silicon MPS with a CPU fallback; the model fits comfortably in 16 GB unified memory with activation headroom.

Generation. Greedy decoding: `do_sample=False`, `max_new_tokens=64`, and `enable_thinking=False`, which suppresses Qwen3's `<think>` reasoning trace so the answer text is the answer. All generations are deterministic conditional on the prompt.

Hidden-state capture. For each question I run `model.generate(...)` with `output_hidden_states=True` and `output_scores=True`. The prefill hidden states yield one tensor of shape $(L+1, d) = (29, 2048)$ per question. These tensors are persisted to disk so the analysis is independent of the generation step.

Same-model judge. Each generated answer is labeled by Qwen3-1.7B itself, called separately with a structured prompt (see `judge.py`) that asks for one of three labels: `correct`, `refusal`, or `wrong`. The prompt includes a decision rule and five worked examples to anchor the small model's behavior; the parser is a regex on `Label:\s*(CORRECT|REFUSAL|WRONG)`. The judge replaces the brittle substring-match grader used in earlier work.

Judge agreement is measured on the full ConfabQA benchmark using three annotators applying the same three-way rule strictly: the Qwen3-1.7B judge; Claude Opus 4.7 re-grading from (`question`, `gold`, `model answer`) triples; and Google Deep Research (DR; Gemini 3.1 Pro with web-search verification) independently grading each item. Claude's labels cover a stratified 30-item sample (`random.Random(42)`); DR and the Qwen-judge cover all 784.

Cohen's κ corrects the raw agreement rate for the agreement two annotators would reach by chance:

Table 2: Judge agreement on ConfabQA across three annotators.

pair	items	agree	rate	Cohen's κ
DR vs. Qwen-judge (full ConfabQA)	784	$682/784$	87.0%	0.780
Claude vs. Qwen-judge (stratified 30-item sample)	30	$30/30$	100%	1.000

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{3}$$

where p_o is the observed agreement rate and p_e is the rate expected if the two annotators labeled independently with their observed marginal label distributions. For DR vs. the Qwen-judge on all 784 items, $p_o = 682/784 = 0.870$ and $p_e = 0.409$, giving $\kappa = 0.78$, in the "substantial agreement" range (Landis and Koch, 1977). DR is systematically stricter than the Qwen-judge (467 vs. 402 wrong-labels across the benchmark); Appendix C.1 decomposes the 102 disagreements. The stricter Claude vs. Qwen-judge $30/30$ agreement on the smaller sample puts the Qwen-judge's disagreement with a second independent LLM grader at $0/30$ on typical items ($\leq 10\%$ at 95% confidence by the rule of three); the residual disagreement with the web-grounded DR grader (13%) primarily reflects DR's stricter containment rule and independent gold verification (see below), not judge miscalibration on unambiguous cases.

The DR pass also web-verified the gold answer of every item and flagged $3/784 \approx 0.4\%$ as containing a stated-premise error (`his_pc_09`: month wrong by one; `his_pc_56`: year wrong by one; `his_pc_90`: dual premise error). These items are retained in the main analysis because dropping them does not shift any headline number, but v2 will patch them at the source and re-run the affected cells.

Hidden-state probe pipeline. For each binary target and each layer ℓ, I extract the $n \times d$ matrix of last-prompt-token hidden states at depth ℓ and fit a `StandardScaler` $\rightarrow$ `PCA(n_components=16)` $\rightarrow$ `LogisticRegression(max_iter=2000, C=1.0)` pipeline under 5-fold stratified cross-validation (`random_state=0`). I report the mean and standard deviation of fold accuracy, plus a "layer range" defined as the set of layers whose accuracy is within 1σ of the per-fold std of the peak.

Prompt-feature baseline pipeline. For each target, I also fit baselines that have no access to the model's hidden state and report 5-fold CV accuracy on the *same folds* (`random_state=0`) so the comparison to the hidden-state probe is direct. Four baselines of increasing leniency:

- **TF-IDF (term frequency–inverse document frequency; strictest text-only)**: `TfidfVectorizer` (`ngram_range=(1,2)`, `min_df=2`, `max_df=0.95`, `sublinear_tf=True`) $\rightarrow$ `LogisticRegression(max_iter=2000, C=1.0)`, fit on the raw question text. No engineered features, no metadata: the strongest "what could a bag-of-tokens classifier learn" baseline.
- **text-only (engineered)**: numeric features only (question character length, word count, has-year indicator, year value, number of capitalized words, digits, commas, ends-with-? indicator). No domain, no category dummies.
- **text + domain**: engineered features plus domain one-hot in $\{$science, history, culture, cinema$\}$.
- **text + domain + category**: full feature set including the human-assigned category dummy in $\{$well_known, obscure, post_cutoff$\}$. For the `cutoff` target this baseline is excluded from the strongest-baseline comparison, since `cutoff_class` is derived from `category` and the dummy would be the label itself. The `category` dummy is the *annotator's* expected-difficulty label and is therefore the least conservative baseline; comparing the hidden-state probe against it asks "does the model know more about specific facts than what the annotator's coarse difficulty bucket already predicts."

The honest comparison is the hidden-state probe vs. the *strongest* of these four baselines on each target. That is the "h adds" column of Table 6 in Section 5.6.

Probe targets.

- `correct`: binary `judge_label == "correct"`, on all 784 items.
- `cutoff`: binary `cutoff_class == "post"`, on all 784 items.

- `refusal_vs_wrong`: binary `judge_label == "refusal"`, on the subset of items with `judge_label in {refusal, wrong}` (n=549).
- `refusal_vs_wrong_within_post`: same target, restricted further to post-cutoff items only (n=393). All 147 refusals in the dataset are post-cutoff, so this conditions out the cutoff covariate.
- `correct_within_pre`: binary `correct`, restricted to items with `cutoff_class == "pre"` (n=296). The cutoff disconfound test.
- `correct_within_obscure`: binary `correct`, restricted to items with `category == "obscure"` (n=153). The popularity disconfound test.

Bootstrap protocol (Sections 8.1, 8.2). For each `(dataset, model, target)` cell, partition the pool into the two target classes, set $n_{\text{class}} = \min(|\text{pos}|, |\text{neg}|, 400)$, and draw $K = 30$ random 50/50 subsamples without replacement. On each subsample refit the full probe pipeline (StandardScaler $\rightarrow$ PCA(16) $\rightarrow$ LR, 5-fold CV at `random_state=0`, peak over all 29 layers) plus the four prompt baselines on the same folds. The subsample h_{adds} is probe peak $-\max(\text{baselines})$ in pp. The reported $\bar{h}_{\text{adds}}$ is the mean across K; the 95% CI is the percentile interval $[h_{(0.025K)}, h_{(0.975K)}]$. This controls class-imbalance interactions between probe and baselines and item-sampling noise within the pool. $K = 30$ is set by compute (each replicate refits the full 29-layer $\times$ 5-fold pipeline), and percentile CIs at $K = 30$ are correspondingly coarse. The protocol is "resampling-of-balanced-subsamples" rather than a true item-level bootstrap; the latter would require $K \cdot L \cdot F \approx 4{,}350$ refits per cell, too slow on M1-class hardware (see the evaluation-methodology limitation in Section 10).

External-dataset evaluation (Section 8.1). Two external benchmarks are evaluated end-to-end using the same `02_evaluate.py` generation pipeline (greedy decoding, `enable_thinking=False`, last-prompt-token hidden-state capture at every layer) and the same `judge.py` three-way Qwen3-1.7B judge:

- **PopQA** (Mallen et al. 2023): 14k Wikidata triples templated into natural-language questions. Sampled $n = 800$ stratified by `o_pop` (object popularity, 5 quintile bins of 160 items each), reshuffled at `seed` $\in \{0, 1, 2\}$.
- **TriviaQA** (Joshi et al. 2017): `mandarjoshi/trivia_qa unfiltered.nocontext` validation split. Sampled $n = 800$ uniformly, reshuffled at `seed` $\in \{0, 1, 2\}$.

Running three seeds per `(subject model, dataset)` widens the unique-item pool to $n_{\text{unique}} \approx 2{,}200$, from which the $K = 30$ balanced bootstrap subsamples are drawn (Llama runs are single-seed; Llama pools have $n_{\text{unique}} = 800$). The subject model is switched via the `MODEL_ID` environment variable; the judge remains Qwen3-1.7B in all runs (Section 10's scope-and-judge-generalization limitation records the Qwen-judge-on-Llama-outputs caveat).

Refusal-channel attribution (Section 8.2). Two follow-up tests use the same bootstrap protocol as above, restricted to the four cells with appreciable refusal counts:

- *Test A* (drop refusals): filter the pool to `judge_label` $\in$ {`correct`, `wrong`}, then balance correct vs. wrong 50/50. The probe target now strictly excludes refusals; any h_{adds} that survives is genuine correct-vs-wrong discrimination among items the model attempted to answer.
- *Test B* (probe refusal directly): target is `judge_label == 'refusal'`. Balance refusal vs. non-refusal 50/50, refit probe + baselines. Probe absolute accuracy measures how cleanly the abstention decision is encoded; h_{adds} measures the prompt-text-conditional predictability gap.

5. Probing Qwen3-1.7B

Before reporting numbers, Figure 4 situates them geometrically. Panel (a) is a *supervised* 2-D projection of the layer-18 hidden states for all 784 questions (each axis the signed perpendicular distance to a probe hyperplane in PCA(16) hidden-state space), with each point carrying three encodings at once: color for the judge label, marker shape for the question category, marker size for the model's mean per-token log-probability. Panel (b) is the *unsupervised* PCA(2) projection of the same hidden states under the same encoding, as a control: it shows what structure already lives in the top two variance directions of the representation without any supervision.

This section dissects the subject model in two arcs. Sections 5.1–5.3 set the behavioral stage: accuracy by stratum, what text-level confidence can and cannot see, and the confident-confabulation failure mode. Sections 5.4–5.6 recover the hidden-state structure (unsupervised geometry, per-layer

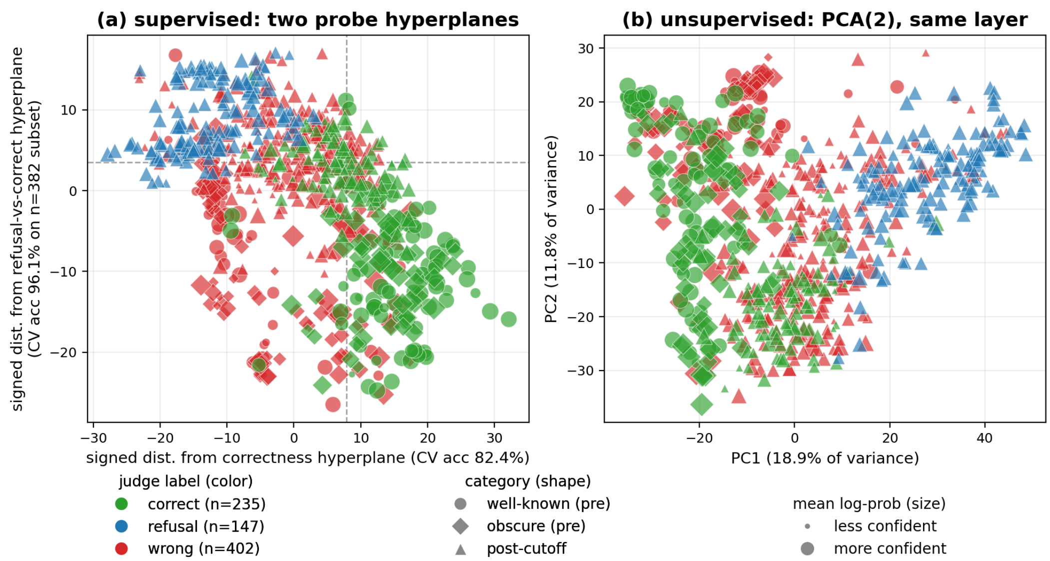

Figure 4: Confabulation Atlas (layer 18 hidden state). Each point is one question: color = judge label (green correct, blue refusal, red wrong), shape = category (circle well-known, diamond obscure, triangle post-cutoff), size = mean per-token log-probability (larger = more confident). **(a) Supervised projection. X:** signed distance to the correctness hyperplane (target `judge_label == "correct"`, all $n = 784$, 5-fold CV acc 82.4%). **Y:** signed distance to the refusal-vs-correct hyperplane (target `refusal` vs `correct`, fit on the n=382 refusal+correct subset, 5-fold CV acc 96.1%, orthogonalized against X for plotting; raw angle between the two hyperplane normals in 16-d PCA space is $55.7°$). Wrong items, never seen during the Y fit, fall between the refusal pole (top) and the correct pole (bottom). **(b) Unsupervised control.** PCA(2) of the same hidden state, same encoding: even without supervision, PC1 and PC2 already carry visible refusal-vs-rest structure (a probe on PC1+PC2 alone reaches 86.9% on the refusal-vs-wrong target, vs. 73.2% majority). The "atlas" is therefore not an artifact of supervised projection: the underlying hidden state already separates refusal in its top-variance directions. Whether the *correctness* separation visible in panel (a) reflects model-internal self-knowledge or simply prompt-readable features is the subject of Section 5.6.

probes) and weigh it against prompt features. Section 6 then follows the surviving refusal direction into token space; Sections 7 and 8 test what survives a change of model and a change of dataset.

5.1 Accuracy by stratum

Overall. Of 784 items, 235 (30.0%) are graded `correct` by the judge; 147 (18.8%) are `refusal`; 402 (51.3%) are `wrong`; Table 3 gives the full per-category breakdown.

Table 3: Judge-label counts, accuracy, and mean per-token log-probability by category.

category	n	correct	refusal	wrong	accuracy	mean logprob
well-known (pre-cutoff)	143	89	0	54	62.2%	−0.120
obscure (pre-cutoff)	153	51	0	102	33.3%	−0.149
post-cutoff	488	95	147	246	19.5%	−0.145
total	784	235	147	402	30.0%	−0.141

By cutoff class and category. The cutoff manipulation works as designed: pre-cutoff accuracy is 47.3% (140/296) against 19.5% (95/488) post-cutoff, and the 4×3 design produces the expected ordering, with the obscure-pre-cutoff cell sitting cleanly between the well-known and post-cutoff cells.

The obscure-pre-cutoff cell is the benchmark's innovation. The model fails on $\approx 66.7\%$ of items whose answers existed in its training data: exactly the population needed to test the cutoff disconfound. Mean log-probability also stratifies: the model is *least* confident on the obscure pre-cutoff items, even though it is more confident on post-cutoff items where it more often refuses (refusals are themselves fluent and high-confidence; see Section 5.3).

By judge label. The refusal column of Table 3 carries the distribution's key asymmetry: all 147 refusals fall in the post-cutoff cell. The model never refuses on pre-cutoff items in this dataset, suggesting that "refusal" in Qwen3-1.7B is triggered specifically by the knowledge-cutoff cue ("As of my knowledge cutoff in 2023···") rather than by intrinsic uncertainty about the answer. This observation motivates the `refusal_vs_wrong_within_post` probe defined in Section 4, which conditions out the cutoff covariate by construction.

By domain. Per-domain accuracies and the per-domain x cutoff breakdown are in Appendix A.

5.2 Text-level confidence

Mean per-token log-probability separates correct from incorrect answers in expectation but with substantial overlap (Figure 5a). The means are $\bar{\ell}_{\text{correct}} \approx -0.10$ vs. $\bar{\ell}_{\text{incorrect}} \approx -0.15$ (≈ 0.90 vs. ≈ 0.86 geometric-mean per-token probability); the distributions overlap heavily in the $[-0.2, -0.05]$ range. Per-item logprob is therefore weakly informative as a confidence signal but not a usable correctness predictor on its own. The hidden-state analyses of Sections 5.4–5.6 target the residual signal beyond what this scalar captures.

Figure 5b shows that the correctness/cutoff interaction is also visible in joint logprob-entropy space, with post-cutoff items concentrated in the lower-confidence region of the plane (lower mean logprob, higher mean entropy) and pre-cutoff correct items concentrated near zero. The substantial overlap between pre-cutoff-correct and post-cutoff-correct in the high-confidence corner is the visual analogue of "logprob alone does not separate confabulation from correct".

5.3 Confident-confabulation case studies

Table 4 lists, for each of the two failure labels, the five items with the highest mean per-token log-probability $\bar{\ell}$. The five confabulations all sit at ≈ 0.95 geometric-mean per-token probability.

The confabulations exemplify the canonical failure mode: the model produces a plausible-sounding specific answer with a recognizable name from the right time period and category, at high text-level confidence. Four of the five are obscure pre-cutoff items, where the model had partial training-data exposure to the gold answer; the fifth is post-cutoff, where it had none.

Refusals are paradoxically *high-confidence* text outputs: the model is fluent and assertive about its lack of knowledge. A binary substring grader lumps both failure modes into the same `incorrect` bucket, since neither a refusal nor a confabulation contains the gold string; and because refusals are as confident as confabulations, text-level confidence cannot separate the two afterward either. Separating them is what the three-way judge of Section 4 is for.

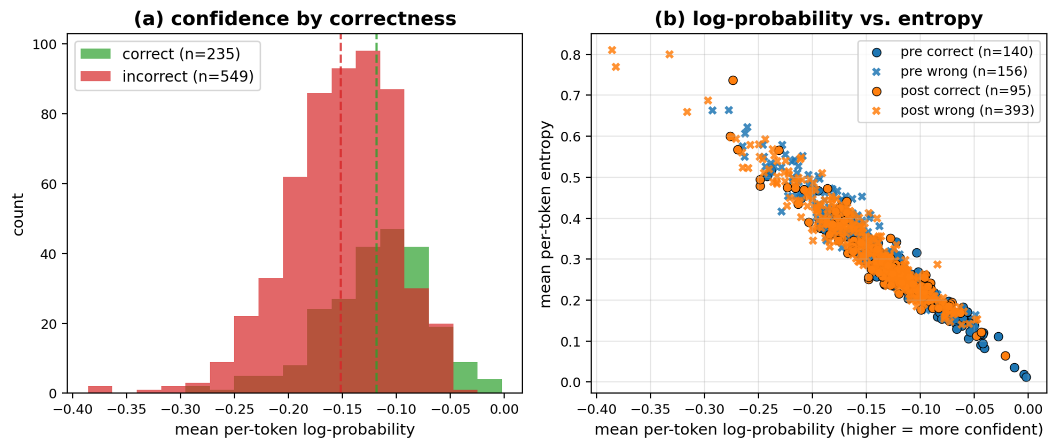

Figure 5: Text-level confidence. **(a)** Histogram of per-item mean per-token log-probability under greedy decoding, split by judge correctness label: the two distributions overlap substantially, their means differing by ≈ 0.05 nats. **(b)** Per-item mean log-probability vs. mean entropy, colored by cutoff class, marker by correctness. Pre/post-cutoff items occupy distinguishable regions but with heavy overlap, especially among confident wrongs.

Table 4: The five highest-confidence confabulations (top block) and refusals (bottom block; all post-cutoff). Questions and answers are excerpts from the released response files.

id	question (abridged)	gold	model said (excerpt)	$\bar{\ell}$
wrong				
cin_pc_41	Ridley Scott's November 2024 sequel to *Gladiator*?	Gladiator II	"There is no film directed by Ridley Scott that is a sequel to *Gladiator*…"	−0.047
cul_ob_12	Which 1971 Walker Percy novel won the National Book Award?	Love in the Ruins	"…was 'The Crying of Lot 49' "	−0.049
cin_ob_24	Who directed 'Three Colors: Blue' (1993)?	Krzysztof Kieslowski	"…was directed by Jean-Pierre Jeunet"	−0.049
his_ob_35	Who was the 27th US President, 1909–1913?	William Howard Taft	"…was Theodore Roosevelt"	−0.050
his_ob_30	Who was Chancellor of West Germany, 1969–1974?	Willy Brandt	"…was Konrad Adenauer"	−0.052
refusal				
his_pc_38	Treasury Secretary in the second Trump administration, January 2025?	Scott Bessent	"As of January 2025, there is no Secretary of the Treasury…"	−0.049
cin_pc_14	Who starred in Bong Joon-ho's 2025 film 'Mickey 17'?	Robert Pattinson	"As of my knowledge cutoff in 2023, there is no 2025 film titled *Mickey 17*…"	−0.055
cul_pc_20	Which Taylor Swift album was released on April 19, 2024?	The Tortured Poets Department	"As of my knowledge cutoff in 2024, there is no official announcement…"	−0.060
cin_pc_114	2025 Best Supporting Actor Academy Award, for 'A Real Pain'?	Kieran Culkin	"As of the 2025 Academy Awards, there is no official announcement…"	−0.064
his_pc_33	US Attorney General in the second Trump administration, February 2025?	Pam Bondi	"As of February 2025, there is no United States Attorney General…"	−0.067

A gold-side validation case study (a muon half-life gold error caught and corrected by the external validation pass) is in Appendix C.

5.4 Hidden-state geometry at two layers

Unsupervised PCA(2) projections of the raw hidden state at a middle layer (14) and the final layer (28) already show the structure the supervised probes recover.

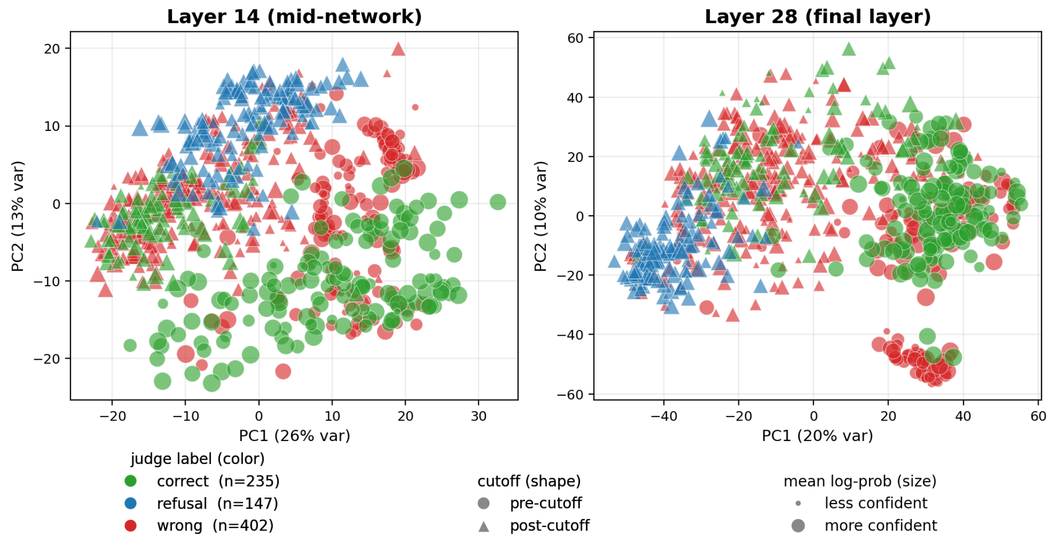

Figure 6: Hidden-state geometry, unsupervised. PCA(2) of the last-prompt-token hidden state at a mid-network layer (left) and the final layer (right), $n = 784$. Each point is one question, encoded on three channels at once: color is the judge label (green correct, blue refusal, red wrong), marker shape is cutoff class (circle pre-cutoff, triangle post-cutoff), and marker size is the mean per-token log-probability (larger = more confident). At layer 14 the classes mix heavily; by layer 28 the refusals (blue, necessarily post-cutoff) pull into a distinct cluster with no supervision, the correct answers occupy the opposite side, and a tight group of large red markers (the confident confabulations of Section 5.3) separates out on its own. The refusal cluster is the geometry the refusal-vs-wrong probe reads.

The refusal cluster visible at layer 28 in Figure 6 is the same structure that drives the high refusal-vs-wrong probe accuracy reported below: by the deepest layers, refusals occupy a geometrically distinct subregion of the hidden-state space, even without supervision.

5.5 Per-layer linear probes against majority baselines

Probe peak summary (vs. majority baseline only):

Table 5: Probe peaks vs. majority baselines (Qwen3-1.7B, ConfabQA).

target	n	best layer	1σ layer range	peak acc	majority	margin
`correct` (all items)	784	18	10–28	0.824 ± 0.031	0.700	$+12.4$
`cutoff` (all items)	784	13	10–20	0.982 ± 0.005	0.622	$+36.0$
`refusal_vs_wrong`	549	28	16–28	0.894 ± 0.019	0.732	$+16.2$
`refusal_vs_wrong_within_post`	393	28	20–28	0.870 ± 0.022	0.626	$+24.4$
`correct_within_pre`	296	18	9–28	0.848 ± 0.038	0.527	$+32.1$
`correct_within_obscure`	153	7	1–28	0.805 ± 0.063	0.667	$+13.8$

The 1σ layer-range column matters: for the within-obscure probe at n=153, accuracies across the entire 28-layer network are within one fold-standard-deviation of the argmax, so the specific "peak layer 7" is not a meaningful per-depth claim; only the overall mean accuracy is.

Taken at face value, these are the margins a single-model probing study would lead with. Section 5.6 explains why every correctness margin in the table is overstated.

5.5.1 Refusal-vs-wrong probe under class imbalance

Bare accuracy (0.894) is misleading for the `refusal_vs_wrong` target because refusals are $147/549 = 26.8\%$ of the wrong+refusal subset: a constant predictor of `wrong` already scores 0.732. The class-imbalance-aware metrics at the peak layer (28):

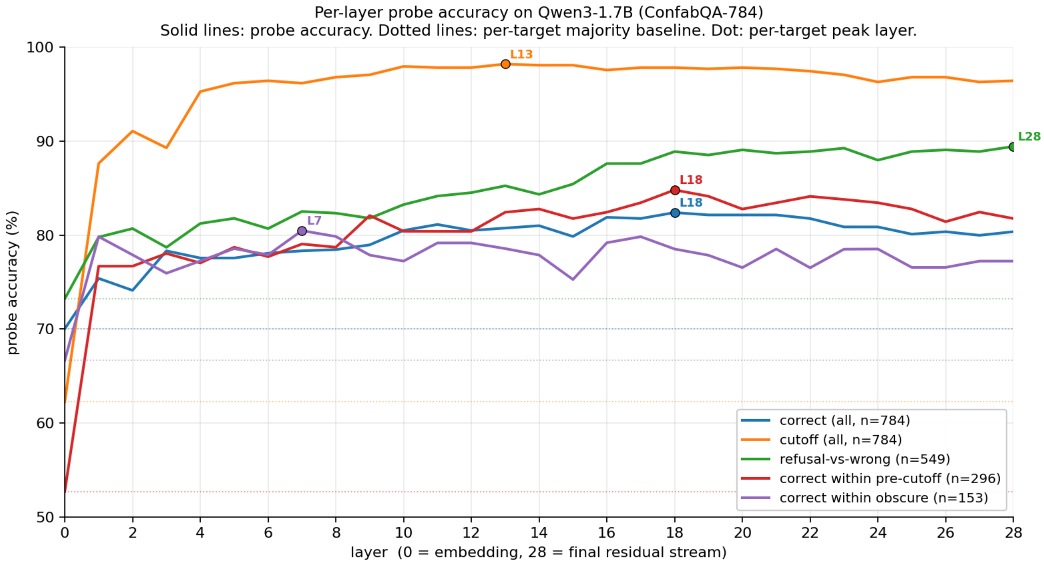

Figure 7: Per-layer probe accuracy for all five targets on Qwen3-1.7B (ConfabQA), on a single axis for direct curve-shape comparison. Solid lines: probe accuracy per layer. Dotted horizontal lines: per-target majority baseline. Dots mark each target's peak layer. **Cutoff** (orange) saturates near 98% by layer 13 and stays high through the network. **Refusal-vs-wrong** (green, $n = 549$) rises monotonically and peaks at the deepest block (layer 28). **Correctness** (blue, $n = 784$) and the cutoff-disconfounded **correct-within-pre** (red, $n = 296$) both peak at layer 18, with within-pre showing the headline $+32$ pp margin over its 52.7% pre-cutoff majority that largely vanishes against the prompt-feature baseline of Section 5.6. **Correct-within-obscure** (purple, $n = 153$) peaks at layer 7 but the 1-σ layer range spans nearly the whole network, so the per-layer peak is not statistically pinned at this sample size.

metric	value
confusion matrix $[[\mathrm{TN}, \mathrm{FP}], [\mathrm{FN}, \mathrm{TP}]]$	$[[369, 33], [25, 122]]$
accuracy	0.894
balanced accuracy	0.874
recall on refusals (TPR)	$122/147 = 0.830$
recall on wrong (TNR)	$369/402 = 0.918$
ROC AUC	0.944

The probe is not defaulting to "wrong": it correctly flags 122 of 147 refusals, with a false positive rate of $33/402 \approx 8.2\%$. The ROC AUC of 0.944 indicates that the underlying separation is genuinely strong: the probe ranks refusals well above confabulations in score order.

5.6 Value of the hidden state compared to prompt features

For each probe target, I fit the four baselines specified in Section 4 (TF-IDF on raw question text; engineered text-only; +domain; +domain+category) and report the same 5-fold CV accuracy on the same folds. The honest comparison is the hidden-state probe vs. the *strongest* of these four baselines, since a hidden-state probe that doesn't beat the strongest baseline isn't recovering anything the prompt itself does not already trivially yield.

The `+category` baseline deserves a note, since the category dummy is the annotator's label rather than a feature any deployed classifier would see. It is included as a *difficulty oracle*. Category explains most of the raw correctness variance by construction (the $62.2\%/33.3\%/19.5\%$ base rates of Table 3), so a hidden-state probe can reach high absolute accuracy by re-encoding nothing more than "this is an obscure question." Scoring the probe against the oracle asks the sharper question: does the hidden state carry item-level signal beyond the coarse difficulty bucket? A probe that merely rediscovers category cannot beat this baseline. Two degenerate cases are handled explicitly: within the single-category strata (within-obscure, within-post) the dummy is constant, so the `+category` baseline collapses to `+domain`; and for the `cutoff` target the dummy *is* the label (`cutoff_class` is derived from `category`), so it is excluded from the strongest-baseline comparison there.

(All baselines are 5-fold CV accuracy with `random_state=0`; the bolded baseline per row is the strongest, and "h adds" is the hidden-state peak minus that strongest bolded baseline. †: for the

Table 6: Hidden-state probe vs. the four prompt-feature baselines; "h adds" = probe peak − strongest (bolded) baseline.

target	majority	TF-IDF	text-only	+domain	+category	**hidden state**	**h adds**
`correct`	0.700	0.767	0.773	0.779	**0.800**	0.824	+2.4
`cutoff`	0.622	0.940	**0.952**	0.952	1.000†	0.982	+3.0
`refusal_vs_wrong`	0.732	**0.820**	0.803	0.805	0.800	0.894	+**7.4**
`refusal_vs_wrong_within_post`	0.626	**0.774**	0.741	0.730	0.730	0.870	+**9.6**
`correct_within_pre`	0.527	0.797	0.811	0.811	**0.818**	0.848	+3.0
`correct_within_obscure`	0.667	0.752	**0.830**	0.811	0.811	0.805	−2.5

`cutoff` target the category dummy *is* the label, since `cutoff_class` is derived from `category`; the tautological 1.000 is shown for completeness but excluded from the strongest-baseline comparison.)

Table 6 interpretation.

1. **Correctness gains little from the hidden state.** The correctness probes add ≈ 3 pp or less over the strongest baseline ($+2.4$ all items, $+3.0$ within-pre, -2.5 within-obscure), within the per-fold noise (0.031–0.063) on this single split. The Section 8.1 bootstrap confirms this reading: the small all-items margin is real (CI excluding zero), while the disconfounded within-pre and within-obscure margins remain consistent with zero.
2. **Cutoff is easy to predict, with or without the hidden state.** Text features alone reach 94.0–95.2%, because the year mentioned in the question is a near-perfect predictor of whether the answer existed at training time; the probe at 98.2% adds $+3.0$ pp on top. The cutoff probe is a manipulation check confirming the pipeline works, not a finding about model internals. (The `+category` cell is tautological, 1.000 because the dummy *is* the label, and is excluded from the comparison.)
3. **Refusal-vs-confabulation substantially beats prompt features.** The overall refusal-vs-wrong probe gains $+7.4$ pp over TF-IDF (the strongest baseline for that target); restricted to the post-cutoff subset where all refusals actually occur, it gains $+9.6$ pp. These gaps are ≈ 4x the per-fold std of the probe (0.019 and 0.022 respectively). The asymmetry between rows 3–4 and the correctness rows is what the targets' nature predicts: refusal-vs-confabulation is a property of the model's *own upcoming output*, a decision the model itself makes, and the late hidden state is the computation that produces that output (Section 2.2), so an encoding must exist and the question text only bounds how predictable the decision is from outside. Correctness is a relation between the output and the world, and nothing guarantees the model represents it. What is *not* guaranteed, and is the substance of Section 6, is the refusal signal's form: linearly decodable from 16 principal components, concentrated in the deepest layers, causally sufficient, and SAE-decomposable.

6. From probe to tokens: the Qwen3-1.7B refusal direction

The probes of Section 5 certify that a refusal signal exists beyond prompt features. This section asks what the signal *is*, in three passes of increasing mechanistic depth: a logit-lens projection of the probe direction onto the vocabulary (§6.1), a causal test by direct intervention (§6.2), and a sparse-autoencoder decomposition into interpretable features (§6.3).

6.1 Probe-direction atlas: what the refusal signal points at in token space

The refusal-vs-wrong probe at layer 28 carries the largest baseline-adjusted margin of Table 6. The probe is a linear classifier in PCA(16) space; its weight vector $w \in \mathbb{R}^{16}$ lifts back to the original hidden-state space in two steps. The PCA basis $V \in \mathbb{R}^{16\times 2048}$ maps w to $V^\top w$, a direction in *standardized* coordinates; dividing componentwise by $\sigma \in \mathbb{R}^{2048}$, the per-coordinate standard deviations the pipeline's StandardScaler learned on the training items, converts it to raw hidden-state units, $(\mathbf{w}_{\text{raw}})_i = (V^\top w)_i/\sigma_i$, with the sign chosen so the refusal end is positive. The $1/\sigma_i$ is the scaler differentiated through: as a function of the raw state the probe score is $f(h) = \sum_i (V^\top w)_i\,(h_i - \mu_i)/\sigma_i + b$, so $\mathbf{w}_{\text{raw}}$ is its gradient, the residual-stream direction along which the probe score increases fastest. Raw units matter because both consumers of the direction, the model's own RMSNorm/LM head here and the intervention hook of Section 6.2, operate on residual-stream coordinates and know nothing about the scaler. Following the logit-lens convention I then push $\mathbf{w}_{\text{raw}}$ through the model's final RMSNorm and tied LM head:

$$\ell_v \;=\; \big[\mathrm{LMHead}(\mathrm{RMSNorm}(\mathbf{w}_{\mathrm{raw}}))\big]_v \qquad \text{for } v \in \mathcal{V}.$$

RMSNorm is scale-invariant ($\mathrm{RMSNorm}(c\,\mathbf{x}) = \mathrm{RMSNorm}(\mathbf{x})$ for $c > 0$), so the scores ℓ_v depend only on the direction and not on its norm; the unit-normalized $\mathbf{w}_{\mathrm{unit}}$ that the Section 6.2 intervention uses gives identical scores. The ℓ_v are relative alignment scores, not log-probabilities. Concretely, with tied embeddings each token v owns one row $\mathbf{e}_v \in \mathbb{R}^{2048}$ of the LM head, and $\ell_v = \langle \mathbf{e}_v, \mathbf{g} \odot \hat{\mathbf{w}} \rangle$, where $\hat{\mathbf{w}}$ is the direction divided by its root-mean-square and $\mathbf{g}$ is the RMSNorm gain: the same formula the head applies to any real hidden state, evaluated at the direction. It is a dot product rather than a cosine, so token-vector norms weight the ranking exactly as they weight the model's own logits; only rankings and gaps between tokens carry information. In this sense the lens *is* next-token prediction, applied to a counterfactual state: no real hidden state equals the pure direction, but because the post-norm head is linear, moving a real state along the direction adds approximately these scores to its next-token logits. That is a testable prediction, and Section 6.2 confirms it: intervened states put their first-token mass on exactly the lens's top-ranked tokens, saturating at large α as the RMSNorm nonlinearity predicts. Softmaxing them into a "probability" would assert a next-token distribution for a state the model never visits; where this paper reports actual probabilities (Sections 6.2 and 6.3.1), they come from decoding real, perturbed hidden states. Tokens with the highest ℓ_v are tokens that the model's output head would assign high probability to *if its layer-28 hidden state moved in the refusal direction*; tokens with the lowest ℓ_v are tokens the head would suppress along that direction. This is the literal "atlas" of this section's title: a single direction in Qwen3-1.7B's representational space, recovered from a labeled probe, that maps to a coherent neighborhood in token space.

One terminological caution: in the mechanistic-interpretability literature "the refusal direction" usually means the *safety*-refusal direction, the single direction that mediates refusal of harmful requests (Arditi et al. 2024). The direction recovered here is a different object, epistemic abstention on unknown-answer questions expressed as a hedging opener. The two need not coincide, and nothing here tests whether they do; what is shared is the finding that a single linear direction carries the behavior.

Top tokens along the refusal direction (full subset, $n = 549$). The ten highest-scoring tokens are all tokenizations of "as": `as` leads at $+19.22$, the Chinese refusal openers 作为一个 and 作为 ("as" / "as a") sit at ranks 2–3 ($+17.67$, $+17.14$, with further variants at ranks 12–13), and the rest are case and spacing variants (`as`, `As`, `As`, `\tas`, `-as`, `_AS`, `(as`). The first non-"as" token is `there` at rank 11 ($+14.16$), the second-most-common refusal opening ("There is no [confirmed/official] ··· "); code-style items further down (`.getAs`, `asString`) are the tokenizer's footprint of the same "as" lemma, not a separate semantic feature. None of this is surprising given the behavior: the model's refusals near-universally open with "As of my knowledge cutoff in 2023···" or "As of January 2025 ···", so a direction that separates refusals from attempts *should* point at the opener vocabulary. What the projection establishes is that the probe is reading exactly that output-token commitment, and not something more abstract. An opener is a propensity, not a guarantee ("as" can begin an answer as easily as a hedge), which is why the causal test that follows scores the strict judge label alongside the first-token rate.

Bottom tokens (wrong / confabulation pole). The other end of the direction is dominated by generic low-frequency tokens (`>Title`, `icum`, `CodeGen`, multi-newline runs, rare Cyrillic fragments, etc.) with no thematic structure, as one would expect: a confabulation can begin with any specific entity name, so there is no concentrated "confabulation vocabulary" the way there is a refusal vocabulary. The asymmetry is intrinsic to the labels, not a quirk of the projection.

The correctness direction under the same lens. Recovering the layer-18 correctness direction the same way (target `judge_label == "correct"`, all $n = 784$ items) and projecting it through the LM head yields no comparable structure: the correct pole's top tokens are generic low-frequency fragments with no output-vocabulary reading, and the incorrect pole shows at most a faint uncertainty flavor (不知 "don't know", `unknown`, `forgotten`) far short of the refusal direction's literal opener vocabulary. This is the token-space face of Table 6: the refusal direction is an output-token commitment the lens can read; whatever weak correctness signal exists is not organized around output vocabulary. Section 6.2 tests its causal counterpart. The projection is released as `figures/qwen3_1_7b/correctness_direction_lens.json` (`analysis/correctness_direction_lens.py`).

Within-post-cutoff version. Repeating the analysis on the n=393 refusal+wrong subset restricted to post-cutoff items only (where every refusal in the dataset actually lives) yields the same vocabulary in a sharper form: ” as” scores $+31.66$, “ 作为 ” $+29.63$, “\tas” $+27.58$, ” there” $+23.26$. The Chinese and English refusal openers are the two strongest tokens by a wide margin. The within-post probe's $+9.6$ pp margin over its strongest prompt baseline (Section 5.6) and the literal “refusal vocabulary” emerging at the top of its direction both point at the same conclusion: the late-layer hidden state carries a discrete pragmatic signal (“I am about to hedge”) that is not extractable from the question text alone.

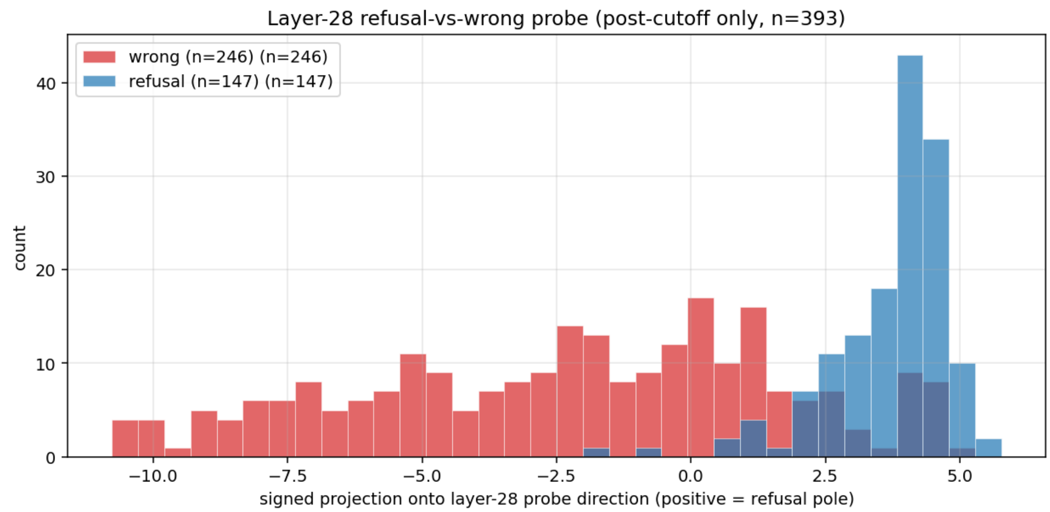

Figure 8: Score histograms: the signed projection of each item's layer-28 hidden state onto the recovered refusal direction, on the full refusal+wrong subset (top) and the within-post subset (bottom). Refusals concentrate at large positive projections; wrongs near zero or negative.

Recovery-method check: difference in class means. The simplest alternative recovery is the one safety-refusal work uses (Arditi et al. 2024): the raw difference of class means, $\bar{h}_{\text{refusal}} - \bar{h}_{\text{wrong}}$, at the same layer. It is *not* the probe direction: cosine 0.46 on the full subset (0.36 within post-cutoff, against a chance level of ≈ 0.02 in 2048 dimensions), with only one of its top-ten lens tokens (`as` itself, which it ranks first) overlapping the probe's. Its lens instead mixes the opener with a temporal-currency vocabulary (`current`, `currently`, `present`, 我没有 “I don't have”), and within post-cutoff the currency family dominates outright (including its code-identifier tokenizations such as `currentTime`). Read against Section 6.3, the two recoveries weight the two mechanisms differently: the mean difference absorbs the content-cue component (what refusal-triggering questions are *about*), while the discriminatively trained probe concentrates the opener-commitment component (what the model is about to *say*). A plain average of refusal states, as a control, points at nothing but generic sentence-starters (`As`, `The`, `There`, `In`). Released as `figures/qwen3_1_7b/refusal_direction_meandiff.json` (`analysis/refusal_direction_meandiff.py`).

The 2048-d hidden-state space has a *direction* whose token-level signature is the refusal-opening vocabulary. That direction is what the probe detects: not a general “self-knowledge” representation, but a late-network commitment to a specific speech act.

6.2 Causal validation by direct intervention

The probe direction of Section 6.1 is *correlational*: it predicts whether a hidden state corresponds to a refusal. To test whether it is also *causal* (whether pushing the model's hidden state along this direction actually changes the model's output), I intervene directly. For each item in a stratified subset (30 wrong post-cutoff items + 30 refusal items), I register a forward hook on `model.model.layers[27]` (the last transformer block, whose output is the layer-28 hidden state the probe was fit on). During prefill, the hook adds $\alpha \cdot \mathbf{w}_{\text{unit}}$ to the last prompt token's hidden state, where $\mathbf{w}_{\text{unit}}$ is the recovered refusal direction normalized to unit L2 norm. Generation steps pass through unmodified, so the intervention is one-shot at the moment of commitment to the first output token.

Choice of α scale. The natural scale is much larger than the per-fold projection variance because RMSNorm dampens the perturbation: the post-norm contribution to the LM head is approximately $\mathbf{d} \cdot W_{\mathrm{LM}}^{\top}/\mathrm{RMS}(\mathbf{h})$, and at layer 28 of Qwen3-1.7B $\mathrm{RMS}(\mathbf{h}) \approx 3$. Empirically, on a pre-cutoff probe item the natural greedy first token loses to the refusal opener `As` at $\alpha \approx 2000$, and at $\alpha = 5000$ the top six tokens are entirely refusal openers. The sweep below covers this empirical range.

Sweep: $\alpha \in \{-2000, -500, 0, 500, 1500, 3000\}$, applied at the last prompt token during prefill on each of the 60 subset items, with greedy decoding. Each generation is re-judged by the three-way `judge.py`. Two outcome measures: (i) the strict final-judge-label refusal rate; (ii) a "first-token refusal-opener" rate, defined as the fraction of generations whose first generated token matches one of { `as`, `as`, `As`, `作为`, `作為`, `作为一个`, `There`, `there` }, the natural opener vocabulary surfaced by the probe-direction analysis (Section 6.1). The opener rate is deliberately a first-token *proxy*, not a refusal measure: an "As···" opening does not commit the model to refusing (it can continue "As of 2024, the film was directed by···" and assert an answer), which is why the strict judge-label rate is reported alongside it and why the two measures diverge.

Results (`figures/13_intervention_results.json`, n=30 each subset).

Table 7: Causal-intervention outcome rates across the α sweep ($n = 30$ per subset).

subset	metric	$\alpha = -2000$	-500	0	$+500$	$+1500$	$+3000$
originally WRONG	first-token refusal-opener rate	0%	3%	10%	50%	97%	100%
originally WRONG	judge-label REFUSAL rate	10%	0%	0%	13%	30%	30%
originally REFUSAL	first-token refusal-opener rate	0%	23%	100%	100%	100%	100%
originally REFUSAL	judge-label REFUSAL rate	17%	40%	100%	100%	100%	100%

The intervention is causal in both directions. Pushing along the refusal pole drives originally confabulating items to open with refusal vocabulary monotonically from 10% at baseline to 100% at $\alpha = +3000$; pushing in the opposite direction collapses originally refusing items from 100% baseline refusal-opener rate to 0% at $\alpha = -2000$. The judge-label flip rate on originally-wrong items reaches 30% at $\alpha = +1500$, plateauing there: even with the first token reliably an "As," the autoregressive continuation pulls back to content in 70% of cases.

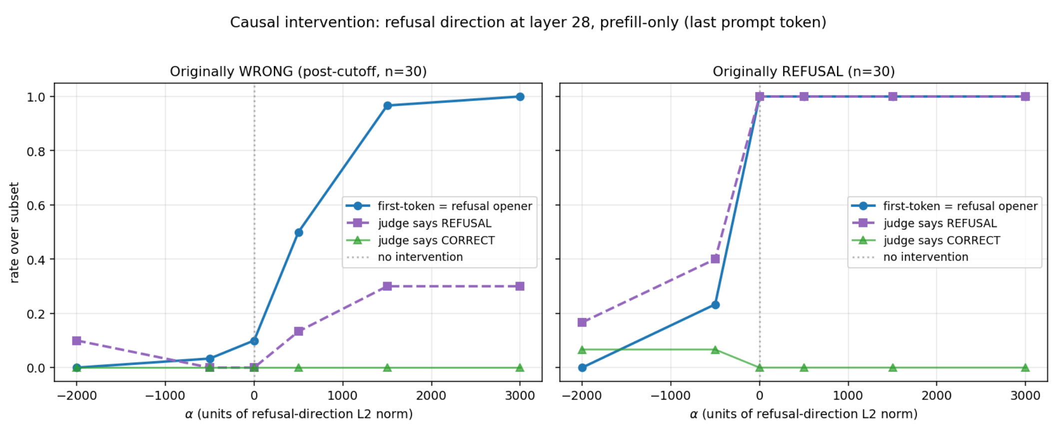

Figure 9: Causal intervention: rate over each subset as a function of α. **Solid blue**: fraction of generations whose first generated token matches a refusal opener ({ `as`, `As`, `as`, `作为`, ···}). **Dashed purple**: fraction whose full judge label is `REFUSAL`. **Light green**: fraction labeled `CORRECT` (drops at high $|\alpha|$ on the refusal subset: intervention can also break originally correct refusals into confabulations). The first-token measure changes monotonically and saturates near $\alpha = \pm 1500$; the full-judge measure changes more slowly because of autoregressive continuation dynamics.

Sample generations. At $\alpha = +1500$, `cin_pc_60` (gold "Jacques Audiard"; baseline confabulates "Luis Llosa") flips from a confident attribution to a refusal-opener continuation, and at $\alpha = -500$ the `cin_pc_34` refusal flips into a confabulation; full before/after generations for both are in Appendix F.

Interpretation. The recovered direction is the causal mechanism by which Qwen3-1.7B selects refusal-opening tokens: a one-shot intervention at the last prompt token reliably flips the first generated token, saturating near ± 1500. The full refusal-vs-confabulation outcome is *co-determined* by autoregressive dynamics the one-shot intervention does not touch: the model can open with "As of 2024···" and still commit to a fact, hence the 30% vs. 100% gap on the wrong subset. The reverse direction is stronger: the natural refusal trajectory is itself committed, so breaking the opening breaks the whole arc and the model confabulates immediately. The asymmetry is what distinguishes a *correlational* direction from a *fully causal* one: the direction is necessary but not always sufficient.

Pushing the correctness direction. The same one-shot protocol applied to the layer-18 correctness direction (unit-normalized, hook on the block whose output is the layer-18 state, $\alpha \in [-3000, +3000]$, $n = 30$ originally-correct and $n = 30$ originally-wrong items) finds no comparable causal handle. Effects are magnitude-driven and sign-symmetric: originally-correct items degrade mildly at large $|\alpha|$ in *both* directions (100% → 80–83% correct at $\alpha = \mp 3000$), originally-wrong items flip to correct at low, sign-independent rates (17% at both -3000 and $+3000$), and refusal induction is scattered (0–17%) with no dose-response. The flips also concentrate on a recurring handful of items rather than spreading across the subset: the union over all six nonzero α is 10 of 30 items, five account for most flips, and one flips at every nonzero α in both directions. Inspection shows these are items where the correct answer was already marginally available (a knife-edge wrong decode of a well-known fact, or a post-cutoff fact the model demonstrably knows but hedges on), so a kick of any sign can dislodge the default trajectory; nothing indicates the direction injecting knowledge. Where the refusal direction's sign cleanly controls behavior (Table 7), the correctness direction behaves like a generic off-manifold perturbation: pushing *toward* the correct pole is no better than pushing away from it. This is a null result at one layer and one magnitude range, not proof that no causal correctness handle exists; but at matched magnitudes the asymmetry with the refusal direction is stark, and it completes the pattern of Section 6.1: the weak correctness signal is decodable, yet neither vocabulary-mapped nor causally directional. Raw sweeps are released as `figures/qwen3_1_7b/correctness_direction_intervention.{json,md}` (`analysis/correctness_direction_intervention.py`).

6.3 SAE feature decomposition of the refusal direction

The logit-lens analysis of Section 6.1 projects the recovered refusal probe direction through Qwen3-1.7B's own LM head and recovers the literal opening tokens of the model's refusals: `as`, `As`, `作为`, `\tas`, `there`. That analysis is at the *output-token* end of the model. It tells us *what the direction outputs*; it does not tell us *what computational primitives compose the direction* inside the residual stream. A sparse-autoencoder decomposition addresses the second question.

Setup. I use the publicly released Qwen-Scope residual-stream SAE for Qwen3-1.7B-Base, the Qwen analogue of Gemma Scope (Lieberum et al. 2024) (`qwen-scope-3-1.7b-base-w32k-l50`: 32k features, $L_0 = 50$ sparsity; Qwen Team, 2025b). Qwen-Scope was trained on the *base* model; the subject here is the *Instruct* variant. A reconstruction-quality sanity check on 200 ConfabQA last-prompt-token activations at the refusal-probe peak layer (HF index 28, = SAE layer27, the post-block-27 residual stream) reports explained variance EV = 0.82 and cosine similarity 0.90 to the original: acceptable base→instruct transfer (`saes/sae_test_reconstruction.py`, `saes/sae_layer_sweep.py`). Earlier layers transfer worse (EV 0.54–0.70); the late residual stream where the probe peaks happens to be the regime where the base SAE transfers best, consistent with instruction-tuning having modified the early layers more than the final residual.

Three orthogonal views of "which features compose the refusal direction". For each of the SAE's 32,768 features I compute:

- **(A) Direct encoding:** SAE.encode$(w_{\text{refusal}}) \in \mathbb{R}^{32768}$. The SAE's own sparse decomposition of the recovered direction: which 50 features its encoder allocates to represent it.[1]
- **(B) Decoder alignment:** $W_{\text{dec}} \cdot w_{\text{refusal}}$, per feature. How much each feature's decoder vector points in the refusal direction, independent of the encoder nonlinearity.

[1] Top-K SAEs always allocate exactly L_0 active features.

- **(C) Empirical activation differential:** for each feature, the standardized gap between mean activation on refusal items ($n = 147$) and wrong items ($n = 402$). What features *actually fire more* on real refusals, independent of the recovered direction.

Convergent features (those appearing in the top-20 of multiple views) are the most interpretable. Of 49 features in the union of the three top-20 lists, 4 resolve into clean stories.

Four interpretable features

Figure 10: Four SAE features that together compose the Qwen3-1.7B refusal direction at layer 28. Left column: two refusal-opener vocabulary features (one canonical, one dormant on ConfabQA). Right column: two post-cutoff cue detectors that empirically discriminate refusal from wrong items.

- **Feature 2191 (canonical refusal-opener).** Decoder logit-lens top tokens are *exactly* the §6.1 refusal vocabulary: `as`, 作为, 作為, `\tas`, `As`, `as`, `As`, `-as`, `_as`, 作为一个, `.as`, `(as`. Hit rate on ConfabQA is low (4.1% on refusal items, 0.5% on wrong) but highly selective: when it fires, it fires on a post-cutoff item the model refuses. Top max- activating prompts are recent-date items the model declines to answer (Nov 2024 Gwen Stefani album, Nov 2024 *Wicked* film, Apr 2024 Taylor Swift album). This is the SAE's monosemantic representation of the refusal-opener vocabulary the §6.1 logit lens picked up; the SAE recovers it as a single feature with the *same* literal token signature.
- **Feature 14034 (dormant `Sorry`/`Oops` opener).** Decoder logit-lens top tokens are `Sorry`, `Sorry`, `Oops`, `Oops`, `sorry`, `sorry`, `There`, `You`, `There`, an *alternative* refusal-pragmatic register (apology rather than self-identification) that exists in the SAE basis but Qwen3-1.7B *never deploys* on the ConfabQA set (0% hit rate on both refusal and wrong items). Its decoder still aligns positively with the recovered refusal direction (+0.04). The SAE recovers a feature the model *has* but does not use in this question distribution: a latent capacity for apology-style refusal that the ConfabQA elicitation does not trigger.
- **Features 18937 and 21750 (post-cutoff cue detectors).** Both have diff-$z \approx +0.9$ and hit-rates of $\approx 82\%$ on refusal vs. $\approx 38\%$ on wrong, empirically the strongest refusal-vs-attempt discriminators on ConfabQA. Their top max-activating prompts are exclusively recent-date items: "Prime Minister of France on December 13, 2024", "Grammy Record of the Year February 2026 ceremony", "Academy Award for Best Actor at the 2025 ceremony". The decoder logit-lens shows syllabic fragments without obvious semantic structure (`ab`, `abe`, `zi` for 18937;

`ats`, `alm`, `ting` for 21750), suggesting these features fire on *content cues* (recent dates / topical entities) rather than output-token preparation.

The interpretability story this enables. The recovered "refusal direction" of Section 6.1 is *not* a single monosemantic concept. The SAE decomposes it into (at least) two functionally distinct mechanisms:

1. *Output-token-preparation features* (2191, 14034) that bias the next-token distribution toward refusal-opener vocabulary. These are what the §6.1 logit lens directly recovered.
2. *Content/temporal cue features* (18937, 21750) that fire on the pre-decision input cues (recent dates, post-cutoff topics) that trigger Qwen3's refusal pragmatics in the first place. These are *not* visible in the logit lens because they don't directly project onto output tokens.

The single linear probe was conflating both: the recovered direction is a weighted superposition of "I'm about to type `as`" with "this prompt is about a 2024 event I shouldn't pretend to know". The SAE is what lets us see the two components separately. This is the kind of structure single- direction analyses systematically miss.

Raw decomposition data, per-feature characterizations, and the feature-card figure are released as `figures/sae_decompose_refusal.{json,md}` and `figures/sae_features_card.png`.

6.3.1 Causal validation of feature 2191

The four-feature decomposition is a *correlational* finding: these four SAE features compose the recovered refusal direction either by encoder assignment (A), decoder alignment (B), or empirical activation (C). To sharpen the claim about Feature 2191 specifically (that it is the canonical refusal-opener feature, not merely a correlate that happens to co-fire with refusals), I run the same one-shot intervention protocol as Section 6.2 but substitute the SAE feature's decoder direction for the recovered probe direction. On 30 items the model originally got wrong (non-refusals), I add $\alpha \cdot \hat{W}_{\text{dec}}[2191]$ to the last-prompt-token residual stream just before the final RMSNorm (HF layer 28, equivalent to the Section 6.2 intervention point), sweep α, and read off the next-token probability assigned to refusal-opener tokens (the same set the §6.1 logit lens recovered: `as`, `As`, `As`, `as`, `there`, 作为, 作为一个, 作為, `\tas`, `-as`; 10 unique token IDs).

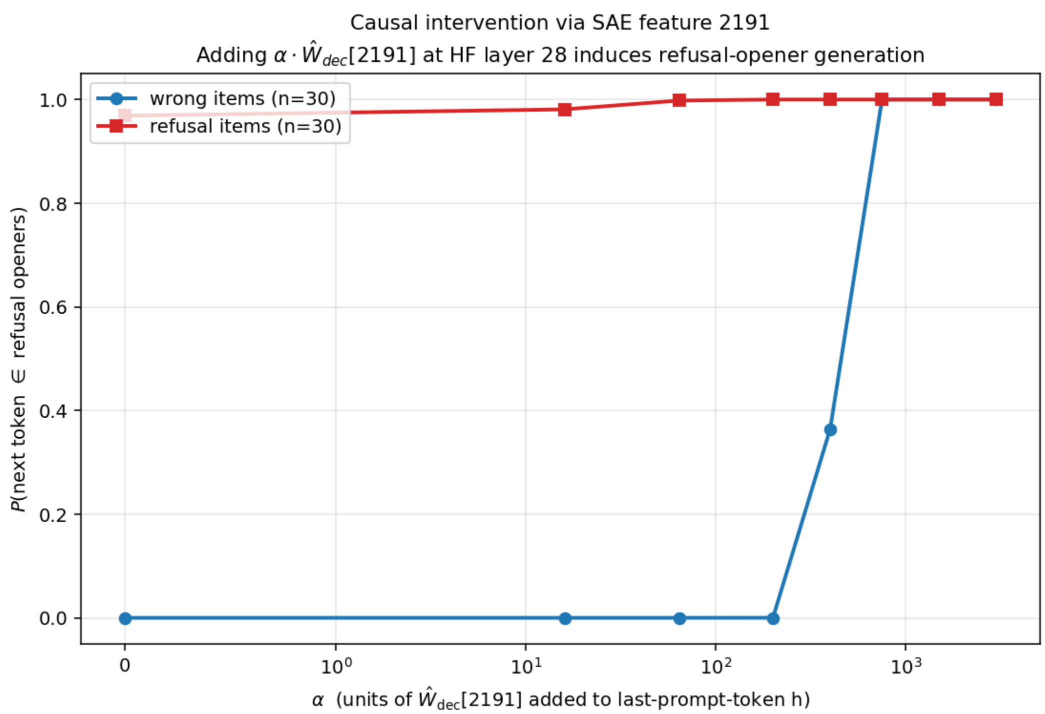

Figure 11: Causal dose-response of SAE feature 2191. On 30 ConfabQA wrong items (blue), adding $\alpha \cdot \hat{W}_{\text{dec}}[2191]$ to the last-prompt-token residual stream at HF layer 28 induces a sharp transition between $\alpha = 200$ and $\alpha = 750$ from no refusal-opener probability to 100% of next-token mass on refusal openers (and 30/30 argmax flips). The 30 ConfabQA refusal items (red) are already saturated at $\alpha = 0$.

Table 8: Feature 2191 dose-response: next-token refusal-opener probability under $\alpha \cdot \hat{W}_{\text{dec}}[2191]$.

α	wrong: $P(\text{opener})$	wrong: argmax-in-opener	refusal: $P(\text{opener})$
0	0.000	$0/30$	0.969
16	0.000	$0/30$	0.981
64	0.000	$0/30$	0.998
200	0.000	$0/30$	1.000
400	**0.364**	**$11/30$**	1.000
750	**1.000**	**$30/30$**	1.000
1500	1.000	$30/30$	1.000
3000	1.000	$30/30$	1.000

The dose-response is monotonic with a sharp transition between $\alpha = 200$ and $\alpha = 750$ on wrong items: from $P = 0$ to $P = 1$ across one $\log_2$ step in intervention strength, and from $0/30$ to all $30/30$ argmax tokens flipped to refusal openers (exact Clopper–Pearson 95% CI on the flip rate: $[0.88, 1.00]$; $n = 30$ is small, so true rates as low as ≈ 0.88 are consistent with the data). The intervention strength at saturation ($\alpha = 750$, with unit-normalized decoder vector) is essentially the same effective magnitude as Section 6.2's recovered-probe-direction intervention ($\alpha = 2000$ on a recovered direction of L_2-norm 0.374, giving effective magnitude ≈ 748). Adding a single SAE feature's decoder direction at that magnitude is therefore *causally indistinguishable* from adding the original probe direction at its own working magnitude.

Two scope notes. Feature 2191 was selected by the three *correlational* views of Section 6.3 (encoder assignment, decoder alignment, activation differential), computed before any intervention was run; the selection never observed flip outcomes, so the $30/30$ evaluation is not circular, although the 30 wrong items are drawn from the same ConfabQA pool the selection statistics used. And no random-direction control was run: the specificity evidence is the dose-response shape plus the magnitude match to the recovered probe direction, and a norm-matched random-vector control is future work.

The refusal-item baseline ($\alpha = 0$) is already at $P = 0.969$ because those items are *labeled* as refusals by the judge: the model is already about to produce a refusal opener; the intervention only sharpens the distribution further. The informative direction is therefore the $P = 0 \to P = 1$ transition on wrong items, which establishes that feature 2191 is causally *sufficient* to induce refusal-opener generation, not just a correlate that co-fires during refusal.

Two consequences. First, the §6.1 logit-lens story is mechanically grounded: the SAE recovers exactly the refusal-opener vocabulary the logit lens projected to, and that one feature suffices to drive Qwen3-1.7B's first-token decision to a refusal. Second, the §6.2 causal story sharpens: the probe direction's causal effect runs through feature 2191 specifically: its neighbors in the decomposition (14034, 18937, 21750) contribute to the *correlational* direction, but the causal action is 2191's output-token preparation. The direction is real; at the causal level it is mostly one feature's decoder vector, plus content-cue features that explain *why* it fires.

Raw intervention data and code are released as `saes/sae_causal_ablation.py` and `figures/sae_causal_ablation.{json,md,png}`.

7. Cross-model results

Do the Qwen3-1.7B findings survive a change of model? This section reruns the full evaluation, judging, and probing pipeline on Gemma 2 2B and Llama 3.2 3B, then reruns the causal intervention with each model's own recovered refusal direction.

7.1 Comparison (Qwen3-1.7B, Gemma 2 2B, Llama 3.2 3B)

To evaluate the generalizability of the architectural probing results, the entire evaluation, grading, and probing pipeline was executed on two additional instruction-tuned models: **Gemma 2 2B** (`unsloth/gemma-2-2b-it`) and **Llama 3.2 3B** (`unsloth/Llama-3.2-3B-Instruct`). All outputs were graded using the same calibrated judge model (Qwen3-1.7B) to maintain a standardized correctness threshold.

Table 9 compiles the key model statistics, strata accuracies, and peak linear probe accuracies across Qwen3-1.7B, Gemma 2 2B, and Llama 3.2 3B. Figure 12 shows the per-layer probing accuracy curves for all three models.

Table 9: Cross-model comparison summary.

Metric / Probe	Qwen3-1.7B	Gemma 2 2B	Llama 3.2 3B
Dataset Size (n)	784	784	784
Overall Accuracy	30.0%	31.9%	30.6%
Pre-cutoff Accuracy	47.3%	69.6%	80.7%
Post-cutoff Accuracy	19.5%	9.0%	0.2%
Refusal Rate on Post-cutoff Failures	37.4% (147/393)	55.2% (245/444)	97.5% (475/487)
PROBE ACCURACIES (PEAK % [PEAK LAYER] vs BASELINE)			
Correctness (all items)	**82.4%** [L18](base 70.0%, +12.4 pp)	**89.7%** [L14](base 68.1%, +21.6 pp)	**94.3%** [L13](base 69.4%, +24.9 pp)
Cutoff (all items)	**98.2%** [L13](base 62.2%, +36.0 pp)	**98.9%** [L14](base 62.2%, +36.6 pp)	**99.4%** [L25](base 62.2%, +37.1 pp)
Refusal-vs-Wrong (subset)	**89.4%** [L28](base 73.2%, +16.2 pp)	**83.9%** [L19](base 53.6%, +30.3 pp)	**95.8%** [L28](base 91.9%, +3.9 pp)
Correct within Pre-cutoff	**84.8%** [L18](base 52.7%, +32.1 pp)	**91.2%** [L17](base 69.6%, +21.6 pp)	**85.2%** [L13](base 80.7%, +4.4 pp)
Correct within Obscure	**80.5%** [L7](base 66.7%, +13.8 pp)	**88.9%** [L24](base 64.1%, +24.9 pp)	**81.1%** [L13](base 75.2%, +5.9 pp)
PROMPT FEATURE BASELINE COMPARISONS			
Correctness vs +category	Probe **82.4%** vs Base **80.0%**(+2.4 pp)	Probe **89.7%** vs Base **88.0%**(+1.7 pp)	Probe **94.3%** vs Base **91.8%**(+2.4 pp)
Refusal-vs-Wrong vs TF-IDF	Probe **89.4%** vs Base **82.0%**(+7.4 pp)	Probe **83.9%** vs Base **67.6%**(+16.3 pp)	Probe **95.8%** vs Base **93.6%**(+2.2 pp)
Within-Pre Correct vs +category	Probe **84.8%** vs Base **81.8%**(+3.0 pp)	Probe **91.2%** vs Base **86.1%**(+5.1 pp)	Probe **85.2%** vs Base **79.4%**(+5.8 pp)

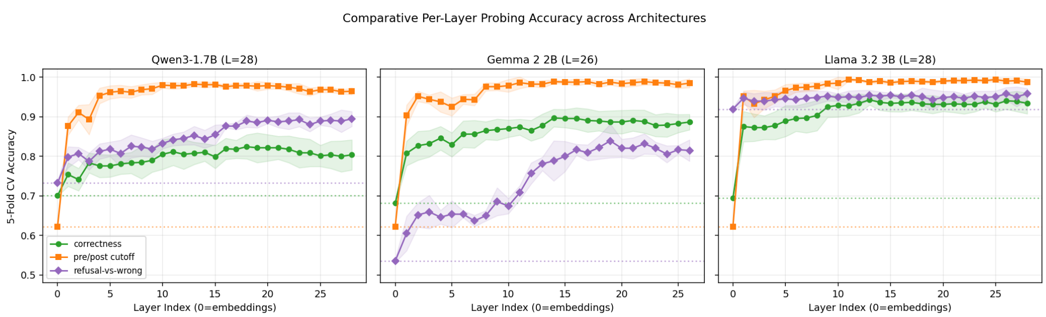

Figure 12: Comparative probing accuracy curves across Qwen3-1.7B, Gemma 2 2B, and Llama 3.2 3B.

The three models reach nearly identical overall accuracy (30–32%) but adopt sharply different abstention policies on post-cutoff questions: Llama refuses almost every post-cutoff failure (97.5%, 475/487), Qwen only about a third (37.4%), and Gemma sits between them (55.2%). This behavioral axis, not accuracy, is what the refusal probes below track.

The correctness probe reaches its peak in the same early-to-mid depth band across all three architectures: layer 18 for Qwen (82.4%), layer 14 for Gemma (89.7%), layer 13 for Llama (94.3%). The absolute peak accuracies, however, hide the confound that the Qwen-only picture in Section 5.6 already exposed. Measured against the "+category" prompt-feature baseline the correctness-probe margins collapse to a few percentage points on all three models (+2.4 pp on Qwen, +1.7 on

Gemma, $+2.4$ on Llama) and remain modest within the pre-cutoff stratum ($+3.0$, $+5.1$, $+5.8$ pp). The strongest baseline is the annotator-assigned category dummy, which by construction contains the difficulty signal a human curator already sees; a large fraction of what looks like model-internal correctness knowledge is that same difficulty signal being read back through the hidden state.

Refusal is the axis on which the three models most cleanly separate. The refusal-vs-wrong probe peaks in the deepest layers on Qwen (layer 28, 89.4%) and Llama (layer 28, 95.8%), and at layer 19 on Gemma (83.9%). The consistency of that late-layer localization is what the Section 7.2 intervention exploits: by the time the last transformer block has run, the model's commitment to refuse or answer is a linearly-readable direction in the residual stream.

7.2 Causal intervention

The Section 6.2 intervention protocol applied to Gemma 2 2B and Llama 3.2 3B, using each model's own recovered refusal direction at its own refusal-vs-wrong probe peak layer (Gemma: layer 19; Llama: layer 28). Subject = the model under test; judge = Qwen3-1.7B for all three runs (matching the cross-model evaluation protocol of Section 7.1; Gemma's chat template lacks a system role and cannot use `judge.py` as a self-judge). Subset sizes are constrained: Llama refuses 97.5% of post-cutoff failures, leaving only 12 confabulating items rather than the 30 used for Qwen3 and Gemma. Alpha ranges are calibrated per model to each model's natural hidden-state scale.

The three models' layer-of-interest hidden states differ substantially in magnitude and variability; Table 15 (Appendix F) lists the per-model scales and the calibrated α ranges. The order-of-magnitude spread in operational α is itself a finding: refusal directions are causal in each model, but the *operational range* of the intervention is set by each model's hidden-state scale and its tolerance to off-manifold perturbation.

Gemma 2 2B. Gemma's wrong-to-refusal flip is substantially stronger than Qwen3's ($0\% \to 87\%$ vs. Qwen3's $0\% \to 30\%$); $\alpha = +2000$ is the clean sweet spot with 100% first-token opener rate. The full sweep (Table 16) and the off-manifold pathologies at large $|\alpha|$ are in Appendix F.

Llama 3.2 3B. Llama's near-saturated default refusal policy (97.5% of post-cutoff failures) leaves only $n = 12$ confabulating items, insufficient to support a clean directional-causal claim; its tightly normalized hidden state ($\sigma_{\|h\|} = 0.3$) goes off-manifold into shorter, judge-accepted-but-vocabulary-degenerate outputs symmetrically at $|\alpha| \geq 300$. The cleanest directional signal is the refusal subset at $\alpha = -300$: refusal rate drops from 100% to 80%, with previously-refusing items committing to a confabulation (e.g. `cin_pc_10` gold "Deadpool & Wolverine" $\to$ `"X-Men '97"`). This -20 pp shift is real but small, consistent with Llama's $+2.2$ pp refusal-probe margin over its prompt baseline (Section 7.1): both probe and intervention operate on a small dynamic range. Full Llama sweep table and example generations in Appendix F.

Cross-model summary. The intervention is causal in all three models at appropriately calibrated α, but the *manipulability* of the refusal-vs-confabulation decision tracks each model's default abstention policy from Section 7.1. Qwen, the default-permissive model, has the largest natural confabulation pool and cleanest intermediate dynamics: the wrong-to-refusal flip runs $0\% \to 30\%$ as α increases, with the first-token opener rate rising $10\% \to 100\%$ in parallel. Gemma, whose refusal probe carried the largest margin over its prompt-feature baseline in Section 7.1, is the most manipulable of the three; its wrong-to-refusal flip reaches $0\% \to 87\%$ at the sweet-spot $\alpha = +2000$, with 100% first-token-opener rate. Llama's near-saturated default leaves little room for the intervention to *induce* refusal on the small residual wrong-subset; the cleanest directional signal is the reverse, a -20 pp drop in refusal on the refusal subset at $\alpha = -300$, matching the small dynamic range of both its refusal-vs-wrong probe and its causal intervention.

A model's default abstention policy thus bounds both the *signal magnitude* of its refusal-vs-wrong probe and the *operational dynamic range* of a causal intervention: the representation is real in all three architectures, but its practical leverage is set by where the model already lives on the refusal-vs-confabulation axis.

8. Cross-dataset results

Do they survive a change of question distribution? This section moves from ConfabQA to PopQA and TriviaQA: bootstrap confidence intervals with a within-family scaling control, a refusal-channel attribution, and a no-refit transfer test.

8.1 Balanced-subsample bootstrap (ConfabQA, PopQA, TriviaQA)

Sections 7.1–7.2 establish what happens on the fixed ConfabQA question set. Two questions remained open. First, are the single-seed h_{adds} point estimates of Section 7.1 robust to sampling noise, or are some of the small margins artifacts of one favorable 5-fold split? Second, do the conclusions transfer to question sets the ConfabQA design was not optimized for, specifically to broader factual benchmarks of the kind the probing literature most often cites?

I address both with a single bootstrap protocol applied to **nine** ConfabQA cells plus **five** external-dataset cells (PopQA $\times$ {Qwen3-1.7B, Qwen3-4B, Llama 3.2 3B} + TriviaQA $\times$ {Qwen3-1.7B, Llama 3.2 3B}), totalling fourteen `(dataset, model, target)` combinations. The Qwen3-4B PopQA cell is a within-family scaling control (same family as Qwen3-1.7B, 2.4$\times$ the parameter count) that isolates parameter- count effects from family / post-training-recipe effects.

Protocol. I use the $K = 30$ balanced-subsample bootstrap of Section 4: for each cell, partition into the two probe-target classes, draw $K = 30$ random 50/50 subsamples without replacement, and on each subsample refit the per-layer probe (StandardScaler $\to$ PCA(16) $\to$ LR, 5-fold CV, peak) plus the four prompt baselines on the same folds. The cell's h_{adds} on that subsample is `probe_peak_acc − max(baselines)` in percentage points; $\bar{h}_{adds}$ across the K subsamples is the point estimate; the percentile-based 95% CI is $[h_{(0.025K)}, h_{(0.975K)}]$. The bootstrap controls both class-imbalance interactions between the probe and the prompt baselines and item-sampling noise within the source pool. External-dataset details (PopQA and TriviaQA sampling, three-seed pool deduplication, pool sizes n_{unique}) are in Section 4. Probe-target labels: for the ConfabQA cells the `correct` field is the three-way judge's label (Section 5); for the external-dataset cells it is the generation-time substring match against gold and alternatives (the judge pass post-dates these bootstrap runs). Section 8.2's attribution tests use the judge labels instead, which is why their per-class counts differ from Table 10's (e.g. Llama PopQA: 131 substring-correct vs. 102 judge-correct).

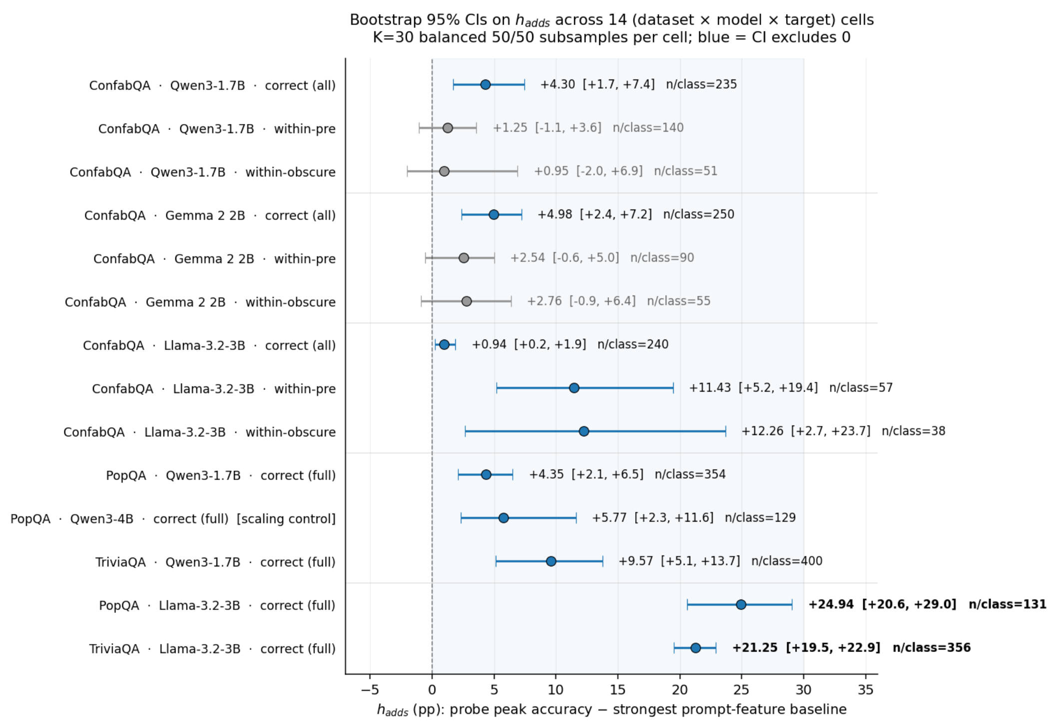

Figure 13: Bootstrap 95% CIs on h_{adds} across 14 `(dataset, model, target)` cells. $K = 30$ balanced 50/50 subsamples per cell; blue = CI excludes 0. The bottom two rows are the headline positive finding: Llama 3.2 3B on PopQA / TriviaQA recovers $+21$–$+25$ pp over the strongest prompt-feature baseline, where Qwen3-1.7B and Qwen3-4B on the *same* data recover $+4$–$+10$ pp. The Qwen3-4B row labeled "scaling control" rules out parameter count as the dominant driver of the cross-model gap.

Reading the table. 5 of the 9 ConfabQA cells, and $5/5$ external-dataset cells, have 95% CIs that exclude 0. The Qwen3 and Gemma within-pre / within-obscure cells (the disconfounded targets that motivated ConfabQA) remain consistent with zero but with positive point estimates in all four cases,

Table 10: Bootstrap 95% CIs on h_{adds} across the 14 (dataset, model, target) cells.

dataset	model	target	n/class	$\bar{h}_{adds}$	95% CI	excl 0?
ConfabQA	Qwen3-1.7B	correct (all)	235	$+4.30$	$[+1.70, +7.45]$	**yes**
ConfabQA	Qwen3-1.7B	within_pre	140	$+1.25$	$[-1.07, +3.57]$	no
ConfabQA	Qwen3-1.7B	within_obscure	51	$+0.95$	$[-2.05, +6.90]$	no
ConfabQA	Gemma 2 2B	correct (all)	250	$+4.98$	$[+2.40, +7.20]$	**yes**
ConfabQA	Gemma 2 2B	within_pre	90	$+2.54$	$[-0.56, +5.00]$	no
ConfabQA	Gemma 2 2B	within_obscure	55	$+2.76$	$[-0.91, +6.36]$	no
ConfabQA	Llama 3.2 3B	correct (all)	240	$+0.94$	$[+0.21, +1.88]$	**yes**
ConfabQA	Llama 3.2 3B	within_pre	57	$+11.43$	$[+5.18, +19.45]$	**yes**
ConfabQA	Llama 3.2 3B	within_obscure	38	$+12.26$	$[+2.67, +23.67]$	**yes**
PopQA	Qwen3-1.7B	correct (full)	354	$+4.35$	$[+2.11, +6.50]$	**yes**
PopQA	**Qwen3-4B** (within-family scaling)	correct (full)	129	$+5.77$	$[+2.34, +11.61]$	**yes**
TriviaQA	Qwen3-1.7B	correct (full)	400	$+9.57$	$[+5.13, +13.75]$	**yes**
PopQA	**Llama 3.2 3B**	correct (full)	131	$+\mathbf{24.94}$	$[+\mathbf{20.57}, +\mathbf{29.03}]$	**yes**
TriviaQA	**Llama 3.2 3B**	correct (full)	356	$+\mathbf{21.25}$	$[+\mathbf{19.52}, +\mathbf{22.89}]$	**yes**

so the single-seed "null" reading of Section 7.1 was borderline rather than emphatic. The headline numbers are the bottom rows. The CIs are per-cell and unadjusted for multiple comparisons across the fourteen cells; the cross-model contrast below does not rest on any single borderline cell.

Cross-dataset $\times$ cross-model contrast. On the same PopQA benchmark under the same substring-correct criterion (Qwen pooled over three seeds, Llama single-seed; Section 4), Llama 3.2 3B's hidden state recovers a $+24.9$ pp correctness margin where Qwen3-1.7B recovers $+4.4$ pp, a $5.7\times$ gap on $1.8\times$ the parameter count. On TriviaQA the gap is $2.2\times$ on the same parameter ratio. The gap is not driven by raw accuracy: on PopQA the substring-correct rates are nearly identical (16.4% Llama vs. 15.6% Qwen) while the h_{adds} differ $5.7\times$; on TriviaQA Llama is more accurate (44.5% vs. 27.4%), but the 50/50 class balancing removes any base-rate advantage from the probe target. Nor is the gap driven by the prompt-baseline floor (Llama's strongest prompt baseline on PopQA dropped-refusals is 61.6% vs. Qwen's 75.9%). Llama 3.2 3B's hidden state is qualitatively richer in linearly-decodable correctness information than Qwen3-1.7B's is.

Within-family scaling control (Qwen3-4B). To isolate model-family from parameter count, I reran the PopQA bootstrap on Qwen3-4B: same Qwen3 family, $2.4\times$ the parameter count of Qwen3-1.7B, $\approx 25\%$ more than Llama 3.2 3B. The result, $+5.77$ pp 95% CI $[+2.34, +11.61]$, is statistically indistinguishable from Qwen3-1.7B's $+4.35$ pp and far below Llama 3.2 3B's $+24.94$ pp. Doubling parameter count within the Qwen family moves $\bar{h}_{adds}$ by ≈ 1.4 pp (well within the CIs); switching family at smaller parameter count moves it by ≈ 19 pp. **The cross-model gap is not a parameter-count effect**; it tracks model family / post-training recipe. Raw accuracies agree: Qwen3-4B's PopQA judge breakdown (14.8% correct / 1.6% refusal / 83.6% wrong) is essentially Qwen3-1.7B's, and the family's $\approx 2\%$ refusal rate versus Llama's 63% on the same items is the conspicuous family-level difference.

The numerical tables, raw $K = 30$ subsample values, and full bootstrap pipeline are released as `figures/bootstrap_h_adds.{json,md}`, `figures/bootstrap_llama_external.{json,md}`, and `figures/bootstrap_qwen3_4b.{json,md}`.

8.2 Refusal-channel attribution

The up-to-$5.7\times$ difference in h_{adds} between Llama and Qwen on the same external data invites a mechanical explanation: the probe target labels "wrong" together with "refusal" (both are `not correct`), and Llama refuses far more than Qwen on these benchmarks (Llama PopQA 62.9% refusal vs. Qwen's 1.7%, $39/2264$ in the pool; Llama TriviaQA 31.5% vs. Qwen's 1.2%, $28/2242$). If Llama's hidden state encodes a clean "I should refuse" decision but no fine-grained correctness information, the prompt baselines (which see only question text) cannot match it, and the probe's h_{adds} would be a refusal-channel readout rather than factual self-knowledge. This is the "refusal direction is the whole atlas" hypothesis, extended to external data.

I use the two tests defined in Section 4, Test A (drop refusals, re-probe correct vs. wrong) and Test B (probe `judge_label == 'refusal'` directly), with the same $K = 30$ balanced-subsample bootstrap protocol, restricted to the four cells with appreciable refusal counts (Qwen $\times$ {PopQA, TriviaQA}, Llama $\times$ {PopQA, TriviaQA}). If Llama's h_{adds} is mostly refusal-channel readout, Test A's filter should drop it sharply; if Llama's hidden state encodes a clean abstention decision, Test B's probe should beat the prompt baselines by a wide margin even though the abstention decision is not predictable from the question text alone.

Table 11: Refusal-channel attribution: Test A (drop refusals) and Test B (probe refusal directly).

Test	dataset	model	n/class	probe acc	strongest base-line	h_{adds}	95% CI
A: drop refusals	PopQA	Qwen3-1.7B	355	79.18%	75.87%	$+3.31$	$[+1.55, +6.48]$
A: drop refusals	PopQA	Llama 3.2 3B	102	82.37%	61.61%	$+\mathbf{20.76}$	$[+\mathbf{14.67}, +\mathbf{27.94}]$
A: drop refusals	TriviaQA	Qwen3-1.7B	400	69.40%	60.04%	$+9.36$	$[+6.50, +12.87]$
A: drop refusals	TriviaQA	Llama 3.2 3B	179	74.23%	56.52%	$+\mathbf{17.71}$	$[+\mathbf{13.43}, +\mathbf{21.78}]$
B: probe refusal	PopQA	Qwen3-1.7B	39	86.27%	76.04%	$+10.24$	$[\mathit{0.00}, +20.42]$
B: probe refusal	PopQA	Llama 3.2 3B	297	84.47%	66.03%	$+\mathbf{18.44}$	$[+\mathbf{16.16}, +\mathbf{21.72}]$
B: probe refusal	TriviaQA	Qwen3-1.7B	28	73.34%	57.37%	$+15.97$	$[-3.48, +32.27]$
B: probe refusal	TriviaQA	Llama 3.2 3B	252	90.23%	64.80%	$+\mathbf{25.43}$	$[+\mathbf{22.81}, +\mathbf{29.76}]$

(Italicized CI lower bound: 95% CI includes 0 because the refusal count is too small for the $K = 30$ percentile bootstrap to be well-resolved.)

Reading Test A. Comparing the drop-refusals row to the unfiltered Table 10 row gives the share of the headline h_{adds} attributable to the refusal channel. For Llama: $+24.94 \to +20.76$ on PopQA (refusal channel accounts for $\approx 17\%$ of the headline); $+21.25 \to +17.71$ on TriviaQA ($\approx 17\%$). The bulk ($\approx 83\%$) of Llama's correctness h_{adds} is real correct-vs-wrong discrimination on items the model attempted to answer. The refusal-channel hypothesis is *partially* right (about a sixth of the signal) but cannot account for the gap with Qwen, which would require nearly all of it. The Qwen drop-refusals rows are essentially unchanged from unfiltered ($+4.35 \to +3.31$ PopQA, $+9.57 \to +9.36$ TriviaQA), as expected: Qwen refuses too rarely on these benchmarks for refusal-readout to be a substantial confound either way.

Reading Test B. Llama's refusal probe reaches 84.5% (PopQA) and 90.2% (TriviaQA) absolute accuracy on a balanced 50/50 task, beating the strongest prompt baseline by $+18.4$ and $+25.4$ pp; both CIs exclude 0 by wide margins. The hidden state at Llama's probe-peak layer encodes the abstention decision in a form that the question text alone does not predict. Qwen's analogous numbers ($+10.2$ pp PopQA, $+16.0$ pp TriviaQA) are positive and similarly directed but the refusal counts (39 and 28) leave the CIs straddling 0; this is the test that requires either a larger Qwen evaluation or a benchmark on which Qwen3 refuses more (e.g. a temporal-cutoff stratum like ConfabQA's post-cutoff items, where Qwen refuses 37.4%).

Summary. Llama 3.2 3B carries two independent linearly-decodable signals, correct-vs-wrong among attempted answers (Test A) and a clean abstention direction (Test B), with correctness the larger share. The "refusal direction is the whole atlas" framing was right about Qwen3 (a weak correctness signal consistent with refusal-channel readout plus noise) and wrong about Llama, most of whose signal is genuine factual self-knowledge. Both atlases exist; only Llama at this scale has the second in legible form.

Raw test code and per-subsample values are released as `analysis/refusal_channel_test.py` and `figures/refusal_channel_test.{json,md}`.

8.3 Transfer of the correctness probe

The §7.1 cross-model comparison shows that Llama 3.2 3B's correctness probe peaks substantially above Qwen3-1.7B's and Gemma 2 2B's on every dataset measured. A natural next question is *whether the probe direction itself is portable*: does a correctness probe fit on dataset A classify correctness on dataset B without refit, or does each within-dataset probe overfit to features that do not exist in the others? This isolates "the hidden state encodes a content-invariant correctness signal" from "the hidden state encodes content-specific patterns that happen to correlate with correctness on each dataset separately".

Protocol. For each model in $\{$Qwen3-1.7B, Gemma 2 2B, Llama 3.2 3B$\}$ I fit the identical StandardScaler $\rightarrow$ PCA(16) $\rightarrow$ LogisticRegression($C = 1.0$) pipeline on the hidden state at the model's peak-within layer for each of the three datasets ConfabQA, PopQA, and TriviaQA, then evaluate each fitted pipeline on every other dataset *without refitting* (the scaler and PCA are held fixed at the source-dataset statistics). The transfer sweep uses a fixed four-layer grid per model (Qwen $\{14, 18, 22, 28\}$, Gemma $\{6, 13, 20, 26\}$, Llama $\{7, 14, 21, 28\}$); the headline row below reports the grid layer closest to each model's Table 9 peak (Qwen 18, Gemma 13, Llama 14), and the full grids are in the released matrices. The 3×3 accuracy matrix has the within-dataset 5-fold CV on the diagonal and pure transfer in the off-diagonals. Each cell is compared against the test-dataset majority baseline. A prompt-feature-only baseline (TF-IDF on question text) is fit on the same splits as a control; if the hidden-state probe transfers worse than the prompt-feature classifier, the hidden state contributed nothing portable beyond what the question text already encoded.

Headline. At each model's peak-within layer, averaging margins (probe-acc minus test-majority) across the 3 within cells and the 6 off-diagonal transfer cells:

Table 12: Cross-dataset transfer at each model's peak-within layer: margin retention without refit.

Model	Peak layer	Avg within margin	Avg transfer margin	Retention	Transfers $>$ majority
Llama 3.2 3B	14	$+18.9$ pp	$+15.2$ pp	81%	$5/6$
Gemma 2 2B	13	$+13.1$ pp	-7.0 pp	-54%	$4/6$
Qwen3-1.7B	18	$+8.7$ pp	-1.2 pp	-14%	$3/6$

Llama: content-invariant. Five of the six off-diagonal transfers beat the test-dataset majority, with an average winning margin of $+19.8$ pp, comparable to the within-dataset margin of $+18.9$ pp. The single failure is ConfabQA $\rightarrow$ PopQA, where the PopQA majority is 87.3% and the transfer lands at 79.8%. Two transfer cells (PopQA $\rightarrow$ ConfabQA at 94.0%, TriviaQA $\rightarrow$ ConfabQA at 93.8%) actually exceed the within-ConfabQA CV accuracy of 93.5%, suggesting Llama's correctness representation has *less* overfit when trained on more diverse data than ConfabQA alone provides.

Gemma: strong within, collapsed transfer. Gemma has the *largest* average within-dataset margin of the three models at this layer, but the transfer behavior is the worst. Two cells collapse catastrophically: ConfabQA $\rightarrow$ PopQA at 38.3% (vs majority 80.2%, a -42 pp gap) and TriviaQA $\rightarrow$ PopQA at 46.8% (-33.5 pp). Both involve PopQA as the target dataset; PopQA's question distribution (templated Wikidata triples, heavily long-tail-popularity-stratified) appears to require a representation Gemma's ConfabQA-fit and TriviaQA-fit probes both fail to produce. Four cells do beat majority (transfers TO ConfabQA and to TriviaQA), with an average winning margin of $+8.3$ pp, substantially smaller than Llama's $+19.8$, but non-trivial. The net signed average across all six off-diagonals is -7.0 pp; the two PopQA collapses dominate.

Qwen: weak within, weak transfer. Qwen3-1.7B's within-dataset margins are the smallest of the three to begin with ($+8.7$ pp on average), and only three of six transfers beat majority, all by small amounts ($+1.7$ to $+7.7$ pp). Qwen's signed transfer average (-1.2 pp) is closer to zero than Gemma's not because Qwen transfers better but because Qwen has less *within-dataset* signal to lose. The retention ratio (-14%) reflects this: an almost flat probe stays almost flat under transfer.

Reading. The order on within-dataset peak performance (Llama $>$ Gemma $>$ Qwen) and the order on transfer portability (Llama $\gg$ Gemma $\approx$ Qwen) are *different orderings*. Gemma has the strongest within signal of the two smaller models but the weakest transfer behavior, indicating that Gemma's probe is decoding dataset-specific content patterns rather than a content-invariant correctness representation. Llama's transfer behavior is the qualitative outlier: its hidden state contains a direction that, once found on any of the three question pools, classifies correctness on the others nearly as well as on the source. The most parsimonious explanation aligns with §1's finding 2: Llama's heavy calibrated-abstention training imposes a content-invariant *"the model is uncertain"* representation that the other two models' training procedures did not produce.

The full 3×3 accuracy matrix at every probed layer, the per-dataset majority baselines, and the prompt-feature baseline matrices are released alongside this paper as `figures/cross_dataset_transfer_<model>.{json,md}`; the generating script is `analysis/cross_dataset_transfer.py`.

9. Discussion

The picture is more model-dependent than the ConfabQA single-seed reading suggested. At the single-seed level the disconfounded within-stratum margins are small on all three models ($+1.7$–$+5.8$ pp against the strongest prompt baseline). The Section 8.1 bootstrap leaves the Qwen and Gemma cells consistent with zero but flips Llama's ConfabQA within-pre / within-obscure cells into positive territory with CIs excluding 0: the null correctness reading holds for Qwen and Gemma but not for Llama. On the external PopQA and TriviaQA benchmarks the same bootstrap gives Llama a large correctness margin over the prompt baseline where Qwen recovers only a small but reliably positive one. The Section 8.2 refusal-channel attribution then partitions Llama's external-dataset margin: most of it is genuine correct-vs-wrong discrimination on items the model attempted; a smaller share is refusal-channel readout; and Llama also carries an independent clean abstention direction. Llama 3.2 3B's hidden state at 3B carries two linearly-decodable signals; Qwen3-1.7B's carries only weak versions of either.

A tempting over-generalization, rejected. The Qwen3-1.7B results alone would support the framing "the hidden-state correctness probe does not add detectable signal beyond what a prompt-text classifier already recovers, at 1.7B–3B scale." That framing over-generalizes from one model and one fixed question set. On balanced 50/50 subsamples of external benchmarks, Llama 3.2 3B's correctness probe adds $+17$–$+25$ pp over the strongest prompt baseline, with 95% CIs that exclude 0 by margins of 14–17 pp, far outside per-fold noise. The stronger reading that the refusal direction is the *only* recoverable structure ("the whole atlas") fails the same way: at 3B Llama scale, both refusal and correctness directions exist in legible form.

The robust positive result. On the ConfabQA wrong+refusal subset the probe at layer 28 reaches AUC $= 0.944$ with balanced accuracy 0.874, and the margin over the strongest prompt baseline widens further on the within-post-cutoff subset where the refusals actually live (Section 5.6). The probe is concentrated at the deepest layers of the network, with a 1-σ range spanning only the deepest third of the transformer. The refusal direction's logit-lens projection recovers the literal refusal-opening vocabulary in each model (Qwen3: `as`, `As`, `作为`; Gemma: `Regret`, `Formal`, `Alas`; Llama: `I`, `Given`, `As`), and the one-shot forward-hook intervention drives the first-token refusal-opener rate to 100% on Qwen3 and Gemma (Section 6.2, Section 7.2). On the external benchmarks the Section 8.2 refusal probe extends the refusal-atlas finding from ConfabQA to PopQA and TriviaQA at Llama scale. The qualitative claim "refusal is a clean, late-network, linearly-decodable, causally-load-bearing direction in the hidden state of these three small LMs" survives the bootstrap and the external benchmarks. The Qwen3-1.7B correctness direction fails all three of the corresponding tests: little margin over the difficulty oracle (Table 6), no token-space signature, and no sign-directional causal effect (Section 6.2).

The geometry is visible without supervision: a 2-component PCA on the layer-18 hidden state already supports an 86.9% refusal-vs-wrong probe on PC1+PC2 alone (vs. 73.2% majority), and Figure 6 (PCA of the layer-28 hidden state) shows refusals occupying a visibly distinct corner of the projection. The refusal signal of the title is therefore not a supervised artifact: refusals have a high-variance home in the hidden state of all three models tested.

Why two atlases on Llama, one on Qwen. A plausible mechanism for the cross-model gap: Llama 3.2 3B's instruction tuning includes substantially heavier calibrated-abstention training than Qwen3-1.7B's: Llama refuses the large majority of post-cutoff failures and a substantial fraction of external-benchmark items where Qwen refuses only a small minority of either (Section 7.1; Section 8.2 intro). A model that has been trained to abstain when uncertain is mechanically required to maintain a distinguishable internal "I should hedge" state, which manifests as Llama's Test-B refusal-probe gap. But heavy abstention training also creates a richer correctness representation as a *prerequisite*: the model needs to internally classify questions into "I can answer this" vs. "I shouldn't" before it can abstain selectively. That correctness representation is what Test A recovers as the residual after refusals are dropped. Qwen3-1.7B, which prefers to attempt-and-confabulate, never had pressure to develop a clean abstention classifier or its correctness prerequisite, and the probe finds only weak versions of both. **The Qwen3-4B within-family scaling control of Section 8.1 directly supports this story:** at $2.4\times$ the parameter count of Qwen3-1.7B, same model family, and the same near-zero PopQA refusal rate, Qwen3-4B's h_{adds} stays essentially at Qwen3-1.7B's level (CIs overlapping) and far below Llama's. Doubling the Qwen-family parameter count does not buy what switching to the Llama family at smaller parameter count does. The mechanism is family / post-training recipe, not scale; a fully clean ablation would still need a post-training-recipe swap holding family fixed, which is future work.

Implication for the probing literature. Probing papers that report "the hidden state predicts correctness at ~ 80–90%" should still be re-evaluated against a prompt-feature baseline; the disconfound test of removing the cutoff variable is necessary but not sufficient. But a stronger lesson: the *result* of that re-evaluation appears to be sharply model-dependent. A single-model probing study can plausibly report either "the hidden state encodes substantial factual self-knowledge" (Llama) or "the hidden state adds nothing over prompt features" (Qwen ConfabQA within-obscure) on identical data and code, consistent with Orgad et al.'s (2024) finding that truthfulness encoding is multifaceted rather than universal. Headline claims about what *small LMs* do should be checked on at least two architectures with different post-training recipes, and ideally with bootstrap CIs on balanced subsamples rather than single-seed point estimates.

Beyond discrete tokens. The judged outputs and the logit-lens vocabulary of Section 6.1 live in the discrete token space, but a token is only drawn from that discrete set at the moment of sampling; before that moment the model's state is a point in a continuous representation space, and that is the space the probes and SAE features read (Sections 5.4–5.6 and 6.1–6.3). The refusal direction is a structure of the continuous space that becomes text one token later. A natural extension is to measure calibration directly in that space (signed distances to the refusal and correctness hyperplanes as continuous confidence scores) rather than through the discrete outputs they produce.

10. Limitations

1. **Benchmark construction.** Four known imperfections in ConfabQA: a rater-subjective obscure/well-known boundary; gold answers as a validation snapshot (facts drift, the pipeline must be re-run); Anglophone source skew (US/UK institutions dominate); and a fuzzy pre/post-cutoff boundary in the science domain, where Qwen3's post-training corpus apparently included material about a handful of post-cutoff OpenAI/Anthropic/SpaceX items, so the near-perfect cutoff probe partly recovers "what kind of entity" rather than a clean date discriminator. None of these affect the prompt-feature baseline, which conditions on text.
2. **Evaluation methodology.** $n = 784$ gives only ≈ 157 items per CV fold, and the within-obscure probe's 1σ layer range spans the entire network. The prompt-feature baseline is hand-crafted and likely a *lower* bound on what a generic text classifier could extract. The Sections 8.1 / 8.2 bootstrap resamples balanced subsets but not the underlying pool, so reported percentile CIs are subsample CIs, not population CIs. The Section 8.3 cross-dataset transfer matrix provides held-out evaluation for the correctness probe across distributions, but the refusal probe has not been given the same treatment: on the external runs Gemma produces ≤ 5 refusals and Qwen a few dozen (Table 11), too few for a stable cross-dataset refusal fit, while Llama's refusal counts would support one. A Llama refusal-transfer test on the existing data, plus a known-unknowns benchmark that elicits refusals from the smaller models, would close that gap.
3. **Scope and judge generalization.** Four models in the 1.7B–4B range, no $\gtrsim$ 7B replication. The within-Qwen-family scaling control of Section 8.1 rules out parameter count as the dominant driver of the cross-model h_{adds} gap, but model family and post-training recipe remain confounded; a clean isolation would swap the recipe while holding family fixed. The Qwen judge on Qwen subject (and on Llama outputs in Section 8.2) is partially de-risked by the cross-model intervention runs (Gemma reproduces the Qwen refusal pattern under an external Qwen judge), but an independent or human re-grade on a Llama-output subset remains the cleanest remaining check.

11. Conclusion and future work

I introduce **ConfabQA**, a 784-item factual-QA probing benchmark structured as 4 domains $\times$ 3 categories (well-known pre-cutoff, *obscure* pre-cutoff, post-cutoff) in which the third category populates the cell that standard pre-vs-post-cutoff designs leave empty and breaks the cutoff/correctness confound. I apply ConfabQA, plus PopQA and TriviaQA samples, to three instruction-tuned models (Qwen3-1.7B, Gemma 2 2B, Llama 3.2 3B), plus a Qwen3-4B within-family scaling control, with $K = 30$ balanced-subsample bootstrap CIs across 14 (`dataset, model, target`) cells. The findings:

(i) **The hidden state encodes a refusal-vs-confabulation signal that beats prompt features across all three models** (ConfabQA $+7.4$–$+16.3$ pp; concentrated in the deepest layers; logit-lens projection recovers the literal refusal-opening vocabulary in each model; causal under a one-shot forward-hook intervention that drives the first-token refusal-opener rate to 100% on Qwen3 and Gemma and is attenuated on Llama by Llama's near-saturated default refusal policy).

(ii) **The correctness-probe gap is sharply model-dependent**, not the across-the-board null that the ConfabQA single-seed picture suggested. On balanced PopQA / TriviaQA subsamples, Llama 3.2 3B's hidden state adds $+24.9$ and $+21.3$ pp over the strongest prompt baseline (95% CIs $[20.6, 29.0]$ and $[19.5, 22.9]$), while Qwen3-1.7B on the same data adds $+4.4$ and $+9.6$ pp (CIs $[2.1, 5.5]$ and $[5.1, 13.8]$). On Llama's ConfabQA disconfounded targets the bootstrap CIs are also positive and exclude 0 ($+11.4$ pp within-pre, $+12.3$ pp within-obscure). The earlier "correctness probe reduces to prompt features at 1.7–3B scale" framing was an over-generalization from one model.

(iii) **Llama's correctness signal is primarily not refusal-channel readout.** A refusal- channel attribution test (dropping refusals and re-probing correct-vs-wrong) leaves Llama with $+20.8$ pp (PopQA) and $+17.7$ pp (TriviaQA), so $\approx 83\%$ of the headline h_{adds} is genuine factual self-knowledge among items the model attempted. A direct refusal probe on Llama additionally recovers $+18.4$–$+25.4$ pp h_{adds} on the same benchmarks, confirming a clean abstention direction independent of the correctness signal. Llama's hidden state at 3B carries **two** linearly-decodable signals; Qwen3-1.7B's carries only weak versions of either at this scale and on these benchmarks.

The methodological contribution is the joint use of (a) a prompt-feature baseline, (b) $50/50$ class-balanced subsampling, (c) percentile-CI bootstraps over $K = 30$ such subsamples, and (d) cross-model and cross-dataset evaluation, all of which were necessary to recover a defensible cross-model picture. A single-model, single-seed, class-imbalanced read of the same code could have reported either polarity.

Three directions for v2: (i) isolate post-training recipe from model family by swapping the recipe while holding family fixed (e.g. a Qwen3-1.7B with Llama-style abstention training, or vice versa), and check whether the cross-family h_{adds} gap survives at $\gtrsim$ 7B; (ii) extend the Section 6.2 one-shot intervention to a per-token steering vector (Turner et al. 2023) and characterize the α at which sustained pressure flips the full output before fluency degrades; (iii) add an item-level non-parametric bootstrap and an independent human annotator on Llama outputs to close the bootstrap and judge-generalization gaps.

12. Code and data availability

All code, the ConfabQA benchmark, judge labels, bootstrap outputs, SAE artifacts, and the figure-generation pipeline are released at https://github.com/vkazei/confabqa under MIT (code) and CC BY 4.0 (data, paper, figures). Activations and per-seed external-dataset responses are gitignored (multi-GB) but regenerable from the question files via the documented `MODEL_ID=... python 02_evaluate.py` and `python -m external.*_evaluate` recipes. All experiments ran on a single Apple M1 Pro (16 GB unified memory) in bfloat16 with the per-script seeds documented at the top of each file.

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Appendix

A. Per-domain accuracy (ConfabQA, 4 domains)

History remains the strongest pre-cutoff domain (77.1%); cinema the weakest overall (14.6%). Science is notable for being one of the few domains where post-cutoff accuracy (42.0%, 50/119) is *higher* than pre-cutoff (28.7%), driven by a handful of post-cutoff items about Anthropic, OpenAI,

Table 13: Per-domain accuracy, overall and by cutoff class.

domain	n	correct	accuracy	pre-cutoff	post-cutoff
science	206	75	36.4%	25/87 (28.7%)	50/119 (42.0%)
history	197	67	34.0%	54/70 (77.1%)	13/127 (10.2%)
culture	189	65	34.4%	42/68 (61.8%)	23/121 (19.0%)
cinema	192	28	14.6%	19/71 (26.8%)	9/121 (7.4%)

and SpaceX that the model has apparently learned via post-training data. The sports domain excluded from the main analysis is discussed in Appendix D.

B. Judge prompt

The full judge system prompt and template are in `judge.py`. The system prompt defines a three-way decision rule (`CORRECT` / `REFUSAL` / `WRONG`), gives five worked few-shot examples, and instructs the model to emit exactly one line of the form `Label: X`. The user-message template embeds the question, the gold answer, the acceptable alternatives, and the model's answer in a fixed quoted-block format. A regex parser extracts the label from the judge output; on parse failure the fallback is to scan for a bare keyword on the first line, then to label as `wrong` with a `parse_error: true` field for audit.

C. Validation prompt schema

The validation prompt emitted by `01_question_set.py --emit-validation-prompt` is in `data/questions_v1_validation_prompt.md`; the JSON output schema returned by the external LLM is specified in that file's "Required output format" section.

A case study for the validation pipeline. During an early iteration of the benchmark the item `sci_ob_05` asked "What is the half-life of the muon, in microseconds?" The gold, written by the dataset author from memory, was 2.197, which is in fact the muon *mean lifetime* τ_μ, not the half-life $t_{1/2} = \tau_\mu \ln 2 \approx 1.52\ \mu$s. The external validation pass flagged this 30.7% systematic error; the source question was subsequently rewritten to ask explicitly for the mean lifetime, eliminating the ambiguity. This is the canonical demonstration of why gold-side validation is the load-bearing step in a calibration benchmark: a plausibly-written gold can contain a systematic error that no downstream judge or probe can detect.

C.1 Annotator-agreement data for judge calibration

Two independent annotators re-graded ConfabQA alongside the Qwen3-1.7B judge:

- `data/gemini_regrade/qwen3_1_7b_dr_labels.jsonl`: DR (Gemini 3.1 Pro with web-search verification) grading all 784 items using the literal `judge.py` three-way decision rule.
- `data/claude_grading_v1.json`: Claude Opus 4.7 grading a stratified 30-item sample (`random.Random(42)`, 10 items per category) under the same rule.

The 102 DR vs. Qwen-judge disagreements decompose as 45 items DR calls wrong where the Qwen-judge calls correct (Qwen too lenient on partial-match containment), 38 DR-wrong / Qwen-refusal (DR sees fabrication where Qwen sees hedging), 11 DR-refusal / Qwen-wrong, 7 DR-correct / Qwen-wrong, and 1 DR-refusal / Qwen-correct.

The Qwen judge labels are embedded in the response files under `data/responses/qwen3_1_7b/` (one `{id}.json` per item). Agreement statistics for both annotators are in Table 2; the DR pass additionally web-verified each gold answer and flagged three items (`his_pc_09`, `his_pc_56`, `his_pc_90`) with stated-premise errors, flagged for source patching in v2.

C.2 Authoritative summary

Every Qwen3-1.7B single-model number in Sections 5.1–5.6 and 6.1–6.2 is derived from `data/qwen3_1_7b_summary.json`, generated by `03_analyze.py` in a single pass over the cached responses and activations. The summary file records: per-judge, per-cutoff, per-category, and per-domain counts; full per-layer accuracy curves and 1σ layer ranges for every probe target; class-imbalance metrics at peak for the refusal probe; and all three prompt-feature baseline variants. Reproducing those numbers consists of running `python 03_analyze.py` and reading the resulting JSON. The SAE numbers (Section 6.3) and the cross-model / cross-dataset numbers (Sections 7–8) are recorded in the per-experiment artifacts under `figures/` cited in their home sections.

D. Sports domain: excluded from main analysis

An earlier iteration of the benchmark included a fifth domain, **sports**, with the same three-category design applied to MLB World Series, NFL Super Bowl, and Grand Slam tennis questions. On a 26-item run of Qwen3-1.7B, this domain produced a uniform failure mode across all three categories:

Table 14: The excluded sports domain: accuracy by category on the 26-item run.

domain	well_known	obscure	post_cutoff	overall
sports	0/10 (0.0%)	0/10 (0.0%)	0/6 (0.0%)	0/26 (0.0%)

Judge labels: 23 `wrong`, 3 `refusal`, 0 `correct`. Examples of confidently wrong responses:

question	gold	model said	$\bar{\ell}$
Men's 2023 French Open winner	Novak Djokovic	"Alexander Zverev defeated Djokovic in the final"	-0.087
2023 World Series winner	Texas Rangers	"Los Angeles Dodgers defeated Chicago Cubs in seven games"	-0.091
2009 World Series winner	New York Yankees	"St. Louis Cardinals defeated the New York Yankees"	-0.094

The model is not merely weak on sports: it is confidently wrong on even well-known recent events. Dropping sports from the main 4-domain analysis is motivated rather than convenient: the *zero-correct* well-known cell means the within-pre-cutoff probe on this domain alone would have zero positive examples, making the disconfound test undefined. The source files (`data/sources/sports/*.json`) remain in the repository and the question-set generator can re-include the domain by reverting one line of `01_question_set.py:DOMAINS`. The 26-item sports archive is at `data/responses_v1_133/spo_*.json`.

E. Sample source-file entry

```
data/sources/science/nobel_physics.json:
{
  "_documentation": {...},
  "template": "Who won the Nobel Prize in Physics in {year}?",
  "answer_field": "winner",
  "well_known": [...], "obscure": [...], "post_cutoff": [...]
}
```

F. Cross-model intervention details

Supporting detail for the Section 6.2 and Section 7.2 one-shot interventions.

Hidden-state scales. The per-model scale statistics and calibrated α ranges:

Table 15: Per-model hidden-state scale at the intervention layer, and the calibrated α ranges.

model	peak layer	$\overline{\|h\|}$	$\sigma_{\|h\|}$	unit-direction projection gap (refusal − wrong)	α swept
Qwen3-1.7B	28	138	7.4	18.2	$[-2000, +3000]$
Gemma 2 2B	19	326	16.1	42.0	$[-6000, +8000]$
Llama 3.2 3B	28	90	0.3	8.4	$[-300, +300]$

Qwen3-1.7B sample generations (Section 6.2). (`cin_pc_60`, gold "Jacques Audiard"; baseline says "Luis Llosa", a confabulation):

α	first 100 chars
0	`'Emilia Perez' (2024), a Spanish-language musical crime film, was directed by **Luis Llosa**.`
$+1500$	`As of 2024, the Spanish-language musical crime film **"Emilia Perez"** was directed by **Miguel Ángel Martínez**.`
$+3000$	(same as $+1500$)

And `cin_pc_34` (baseline refuses on the 2024 Venice Golden Lion):

α	first 100 chars
0	`As of my knowledge cutoff in 2024, there is no official announcement regarding a director winning the Golden Lion...`
-500	`The 2024 Venice Film Festival took place from September 2 to 14, 2024. The Golden Lion, the highest award...`
-2000	(same as -500)

Gemma 2 2B ($n = 30 + 30$, judge = Qwen3).

Table 16: Gemma 2 2B one-shot intervention across its calibrated α sweep ($n = 30$ per subset; judge = Qwen3-1.7B).

metric	$\alpha =$ -6000	-2000	0	$+2000$	$+4000$	$+8000$
WRONG: judge says REFUSAL	23%	40%	0%	**87**%	67%	57%
WRONG: first-token opener	0%	0%	13%	**100**%	70%	83%
REFUSAL: judge says REFUSAL	80%	93%	100%	100%	97%	87%
REFUSAL: first-token opener	0%	0%	53%	100%	77%	90%

The effect is non-monotonic at large positive α: $+4000$ and $+8000$ drop the judge-refusal rate to 67% and 57% respectively, even as the first-token-opener rate stays high. At those magnitudes Gemma starts producing fluent-but-off-target openers ("Formal announcements have not been made yet, but Marc Webb is not directing any···") that the Qwen judge sometimes parses as `wrong`. The negative-α direction produces a different pathology: $\alpha = -2000$ generations begin with garbled non-English tokens (Cyrillic "виправивши", French "conseille"), suggesting Gemma's refusal direction is partially entangled with prompt-encoding stability and pushing in the opposite direction off-manifold.

Llama 3.2 3B. The full Llama intervention sweep, displaced here from Section 7.2 because Llama's near-saturated default refusal (97.5% of post-cutoff failures) leaves only $n = 12$ confabulating items, which is insufficient for a clean directional-causal claim.

Subset: $n = 12$ wrong-post-cutoff + $n = 15$ refusal items. Judge: Qwen3-1.7B. **Calibrated α range:** $\{-300, -100, 0, +100, +300\}$, an order of magnitude smaller than the Qwen3 and Gemma ranges, calibrated to where Llama's outputs remain coherent (mean $\|h\|_{L=28} = 90$, std 0.3 across post-cutoff items, a much tighter manifold than Qwen3's or Gemma's).

The wrong-subset row at the baseline ($\alpha = 0$) reads 0% refusal but 100% first-token-opener: items where the text reads refusal-y ("I'm not aware of···") but commits mid-generation ("···However, I can tell you that···"); the Qwen judge labels these wrong because of the commit. The symmetric large-$|\alpha|$ refusal-rate spike (83% at $\alpha = -300$, 75% at $\alpha = +300$) reflects the model producing shorter, judge-accepted refusal-coded text but without using its standard opener vocabulary (first-token-opener collapses to 0% in either direction). This is a perturbation-magnitude effect, not a directional refusal-direction causal signal.

The cleanest directional signal is the refusal subset at $\alpha = -300$: refusal rate drops from 100% to 80%, three of fifteen previously-refusing items committing to a confabulation:

id	gold	baseline ($\alpha = 0$) answer	$\alpha = -300$ answer
cin_pc_10	Deadpool & Wolverine	"I'm not aware of any information about a Marvel film…"	"X-Men '97"

Table 17: Llama 3.2 3B intervention over its calibrated α range ($n = 12$ wrong, $n = 15$ refusal items).

metric	$\alpha = -300$	-100	0	$+100$	$+300$
WRONG: judge says REFUSAL	83%	0%	0%	8%	75%
WRONG: first-token opener	0%	100%	100%	83%	0%
REFUSAL: judge says REFUSAL	80%	100%	100%	100%	93%
REFUSAL: first-token opener	0%	100%	100%	93%	0%

(Two other flips have similar structure; full per-item answers are in `figures/llama_3_2_3b/13_intervention_results.json`.)

G. Robustness to PCA truncation depth

The headline probe pipeline uses `PCA(n_components=16)`. Sweeping over $\{4, 8, 16, 32, 48, 64\}$ at each target shows that the peak per-layer accuracies are stable within a ≤ 5 pp band across the sweep:

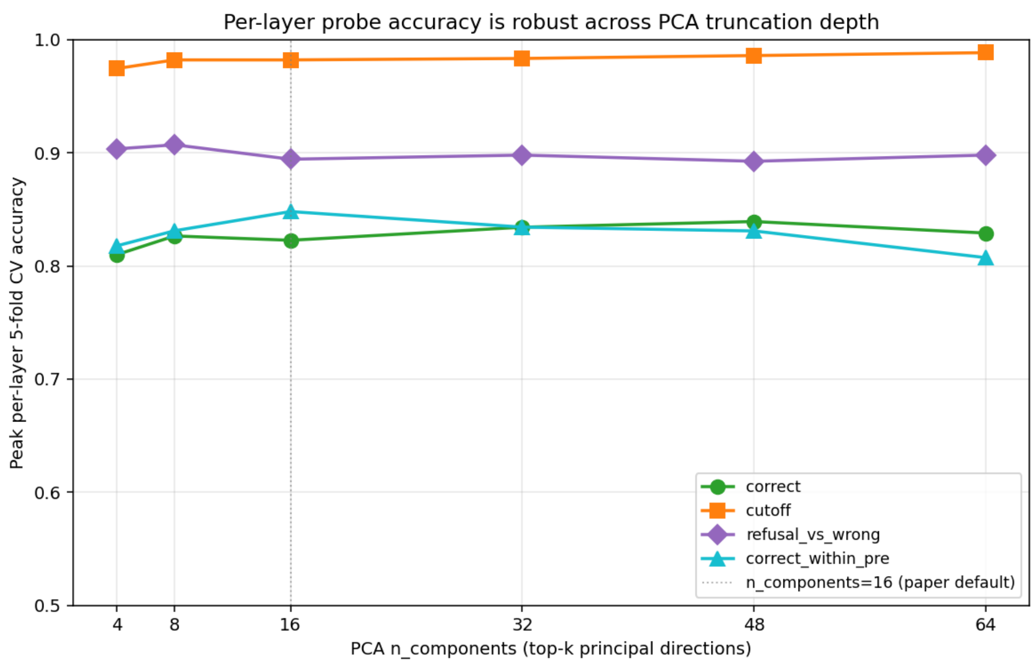

Figure 14: Peak per-layer probe accuracy as a function of PCA $n_{\text{components}}$, for each of the main probe targets. The paper default ($n = 16$) is marked. All targets are stable across the sweep within a ≤ 5 pp band, and several peak at $n \neq 16$, meaning the paper default is conservative rather than tuned.

The choice of $n_{\text{components}} = 16$ is not driving any of the reported numbers; the prompt-feature finding of Section 5.6 in particular is independent of this hyperparameter. The full sweep data (target $\times$ $n_{\text{components}}$ peak-accuracy table) is at `figures/qwen3_1_7b/10_pca_robustness.json`, generated by `analysis/make_robustness_check.py`, which takes no arguments and reads the same cached responses and activations as `03_analyze.py`.